Stationary Reweighting Yields Local Convergence of Soft Fitted Q-Iteration

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Abstract

Fitted Q-iteration (FQI) and soft FQI are widely used value-based methods for offline reinforcement learning, but their standard stability guarantees often depend on Bellman completeness, a strong closure condition that can fail under function approximation. We analyze soft FQI without Bellman completeness and identify the stability mechanism that replaces it: local stationary norm alignment. Near the soft-optimal fixed point, the soft Bellman operator has the same first-order behavior as the policy-evaluation operator for the soft-optimal policy. This operator contracts in the policy's stationary state-action norm, whereas standard fitted regression projects Bellman targets in the behavior norm. This mismatch explains instability under distribution shift. We use this insight to develop stationary-reweighted soft FQI, which reweights each regression step toward the stationary distribution of the current softmax policy. Under approximate realizability and controlled weighting error, we prove finite-sample local linear convergence to the projected fixed point, separating statistical error from geometrically damped weight-estimation error. Our results also show that ordinary soft FQI is locally stable under on-policy stationary sampling, even without Bellman completeness, and explain temperature annealing as a continuation strategy for reaching a contraction region.

1 Introduction

A core objective in reinforcement learning is to compute the optimal value function, defined as the fixed point of a Bellman optimality equation [8]. The hard $\max$ in this operator introduces nondifferentiability and can exacerbate instability when combined with function approximation and bootstrapping. Entropy regularization replaces the hard maximum with a smooth softmax, yielding a differentiable Bellman operator that underlies many modern value-based algorithms [91, 90, 27, 28].

Fitted Q-iteration and its entropy-regularized analogue, *soft* FQI, approximate this fixed point by iterating projected Bellman updates: each step forms bootstrapped one-step targets and regresses them onto a function class. These methods are central in offline RL and large-scale approximate dynamic programming [19, 47, 37, 69], because they are simple to implement and compatible with off-the-shelf supervised learners [77, 24, 38, 2]. When the function class is Bellman complete, standard analyses give finite-sample guarantees for the projected Bellman iteration [26, 66, 48, 21, 61, 20].

However, fitted value-iteration methods can degrade under function approximation and distribution shift when Bellman completeness fails [14, 54, 72]. Information-theoretic, hardness, and classical counterexample results show that, without additional structure, offline value-function approximation can suffer poor horizon dependence, exponential error amplification, or loss of convergence [14, 23, 80, 4, 46, 26, 6, 66]. These limitations have motivated minimax and adversarial Q-learning methods, which relax Bellman completeness through saddle-point formulations [15, 67–69, 32, 81]. However, they introduce more complex objectives, auxiliary critic classes, and delicate optimization and tuning, moving away from the simplicity of supervised learning.

In this paper, we revisit standard soft FQI and ask when it can remain stable without Bellman completeness. We show that stability is governed by a local norm mismatch. Near the soft-optimal solution, the projected soft Bellman update is locally contractive in the stationary norm of the soft-optimal policy, whereas standard soft FQI projects in the behavior-distribution norm. Under function approximation and distribution shift, this mismatch can destroy local contraction. We therefore propose stationary reweighting: a simple reweighting of the least-squares step that targets the stabilizing stationary norm while retaining the supervised learning structure of FQI.

1.1 Contributions

Our main contributions are:

1. **Local contraction of projected soft value iteration.** Near the soft-optimal solution, the soft Bellman operator has the same first-order behavior as policy evaluation under the soft-optimal policy. This identifies the policy's stationary distribution as the natural projection norm. Under this projection, the fitted soft Bellman update is locally contractive and converges linearly, even without Bellman completeness.
2. **Stationary reweighting for off-policy data.** We approximate the stationary-norm projection from behavior data by reweighting each Bellman regression with estimated stationary state-action density ratios. Exact ratios are unnecessary; the weights only need to preserve the stabilizing projection norm up to controlled error.
3. **Finite-sample local convergence guarantees without Bellman completeness.** We prove finite-sample local convergence for stationary-reweighted soft FQI. Inside a contraction basin, the iterates converge linearly up to statistical and weight-estimation errors, with weight errors damped geometrically across iterations. As a special case, ordinary unweighted soft FQI is locally stable under on-policy stationary sampling, even without Bellman completeness.

Scope. Our guarantees are local: they show that stationary reweighting stabilizes soft FQI within a contraction basin of the population projected update. We do not prove basin entry from arbitrary offline initializations, and we condition on sufficiently accurate stationary-ratio weights. Section 4 and Appendix H.4 discuss warm starts and temperature annealing as one possible basin-entry strategy; Appendix H.3 discusses estimation of stationary ratios using DICE-style, minimax, and emphatic occupancy-ratio methods [41, 49, 88, 67, 62, 30].

1.2 Related Work

Several approaches address instability of fitted Bellman updates under function approximation and distribution shift. One line strengthens the approximation structure through linear or low-rank assumptions, representation learning, or state-space discretization [44, 18, 84, 33, 13, 55, 76, 81]. Classical FQE/FQI analyses typically handle distribution shift through concentrability or distribution-mismatch constants [48, 61, 20, 83]. Another line replaces least-squares Bellman regression with minimax, adversarial, or pessimistic Bellman-error objectives [67, 32, 81, 69, 82]. These methods test Bellman residuals against auxiliary critic classes and obtain guarantees under conditions such as partial coverage, dual realizability, or critic richness. Related entropy-regularized control methods [91, 28, 25, 51, 87, 36] optimize policies directly rather than through regression-based fixed-point iteration. In contrast, we keep the fitted soft Q-iteration template and ask whether changing only the regression norm can restore local contraction.

Weighting methods play several related roles in reinforcement learning. Importance sampling reweights returns or trajectories for off-policy evaluation [56, 57, 63, 16], while density-ratio and DICE-style methods estimate stationary or discounted occupancy corrections for off-policy evaluation and learning [41, 29, 49, 88, 67]. Emphatic weighting is closest in motivation: it reweights Bellman-style TD recursions to mitigate off-policy bootstrapping instability, mainly in policy evaluation [85, 42, 62, 30, 86, 54]. Stable ratio estimation under nonlinear approximation and limited coverage remains difficult, motivating regularization, variance control, and truncation [89, 43, 68].

Our setting differs from this literature in two ways. First, we study fitted soft Q-iteration, where weights define the projection norm in a least-squares Bellman regression rather than update weights in a TD recursion. Second, the relevant ratio is induced by the optimal policy and is therefore coupled to the Bellman fixed point. For policy evaluation, prior work has shown that stationary ratios can stabilize Bellman-based methods, including TD learning and FQE [76, 54, 72]. We show that, in soft control, the optimal stationary ratio defines the local projection norm for soft FQI, yielding local contraction near the soft-optimal fixed point without Bellman completeness or global contraction of the projected operator.

2 Preliminaries

2.1 Soft Optimal Control

We consider a discounted MDP with continuous state space $\mathcal{S}$, finite action space $\mathcal{A}$, transition kernel P, reward r_0, discount $\gamma \in [0, 1)$, and behavior distribution ν_b over $\mathcal{S} \times \mathcal{A}$. For any $Q : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ and temperature $\tau > 0$, define the softmax policy

$$\pi_Q(a \mid s) = \frac{\exp\{Q(s, a)/\tau\}}{\sum_{b\in\mathcal{A}} \exp\{Q(s, b)/\tau\}}.$$

The **soft Bellman optimality operator** [27, 28, 69] is

$$(\mathcal{T}Q)(s, a) = r_0(s, a) + \gamma\, \mathbb{E}_{S'\sim P(\cdot|s,a)} \left[\tau \log \sum_{a'\in\mathcal{A}} e^{Q(S',a')/\tau}\right].$$

The **soft optimal Q-function** $Q^\star$ is the unique bounded fixed point of $\mathcal{T}$, equivalently the Q-function maximizing the entropy-regularized discounted return. The **soft optimal policy** is $\pi^\star := \pi_{Q^\star}$.

Our local analysis uses the policy-evaluation operator obtained by freezing the softmax policy at Q. Let $\mathsf{H}(\pi_Q(\cdot \mid S'))$ denote the Shannon entropy of π_Q at S', and define

$$\begin{aligned}
\tilde{r}_Q(s, a) &:= r_0(s, a) + \gamma\, \mathbb{E}_{S'\sim P(\cdot|s,a)} \big[\tau\, \mathsf{H}(\pi_Q(\cdot \mid S'))\big],\\
\mathcal{T}^{\text{eval}}_Q f(s, a) &:= \tilde{r}_Q(s, a) + \gamma(P^{\text{eval}}_{\pi_Q} f)(s, a),\\
(P^{\text{eval}}_{\pi_Q} f)(s, a) &:= \mathbb{E}_{S'\sim P(\cdot|s,a),\, A'\sim\pi_Q(\cdot|S')} \big[f(S', A')\big].
\end{aligned} \tag{104}$$

The entropy-regularized **Bellman evaluation operator** for the soft-optimal policy is $\mathcal{T}^{\text{eval}}_{Q^\star}$.

Let $\mu^\star := \mu_{\pi^\star}$ denote the **stationary state–action distribution** induced by $(P, \pi^\star)$. Let $L^2(\mu^\star)$ be the Hilbert space of functions $f : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ satisfying $\mathbb{E}_{\mu^\star}[f(S, A)^2] < \infty$, equipped with norm

$$\|f\|_{2,\mu^\star} := \big(\mathbb{E}_{\mu^\star}[f(S, A)^2]\big)^{1/2}.$$

Under the stationary norm, $\mathcal{T}^{\text{eval}}_{Q^\star}$ is a γ-contraction [10, 30, 54, 72]. For any nonnegative weight d, write $d \cdot \nu_b$ for the distribution with ν_b-density d; in particular, $(d\mu^\star/d\nu_b) \cdot \nu_b = \mu^\star$.

2.2 Stationary reweighting: intuition and population target

The standard convergence argument for soft value iteration relies on the soft Bellman operator $\mathcal{T}$ being a γ-contraction in the sup norm: iterating $\mathcal{T}$ from any bounded initialization converges to its unique fixed point [8, 58, 10, 25]. With function approximation, fitted Q-iteration does not iterate $\mathcal{T}$ directly. Instead, each Bellman update is projected back onto the function class $\mathcal{F}$, typically by least squares in the behavior norm $L^2(\nu_b)$ [48]. If $\mathcal{F}$ is Bellman complete, so that $\mathcal{T}(\mathcal{F}) \subseteq \mathcal{F}$, this projection is harmless. Without completeness, however, FQI iterates the projected operator $\Pi_{\nu_b}\mathcal{T}$. The difficulty is that Π_{ν_b} is nonexpansive only in $L^2(\nu_b)$, not in the sup norm. Thus $\Pi_{\nu_b}\mathcal{T}$ need not inherit the γ-contraction of $\mathcal{T}$: projection can distort Bellman errors rather than contract them, breaking the stability argument for exact value iteration [26, 66, 14].

The issue is not projection itself, but projection in a norm misaligned with the local Bellman dynamics. Near the soft-optimal fixed point $Q^\star$, the soft Bellman operator is first-order equivalent to policy evaluation under the soft-optimal policy $\pi^\star$:

$$\mathcal{T}(Q) \;\approx\; Q^\star + \gamma\, P^{\text{eval}}_{\pi^\star}(Q - Q^\star) \;=\; \mathcal{T}^{\text{eval}}_{Q^\star}(Q). \tag{1}$$

This follows from the Fréchet derivative identity $D\mathcal{T}(Q)[H] = \gamma P^{\text{eval}}_{\pi_Q} H$, stated formally in Lemma 4. Since $P^{\text{eval}}_{\pi^\star}$ is nonexpansive in $L^2(\mu^\star)$, the linearized update is a γ-contraction in the stationary norm [54, 72]. Thus, the natural local projection norm is $L^2(\mu^\star)$, not the behavior norm $L^2(\nu_b)$.

This motivates replacing the behavior-norm projection with a stationary-norm projection. Assume $\mu^\star \ll \nu_b$, and let $d^\star := d\mu^\star/d\nu_b$. The population stationary-weighted update is

$$Q^{(k+1)} = \arg\min_{f\in\mathcal{F}} E_{\nu_b}\left[d^\star(S,A)\{\mathcal{T}(Q^{(k)})(S,A) - f(S,A)\}^2\right] = \Pi_{\mathcal{F}}\mathcal{T}(Q^{(k)}), \tag{2}$$

where $\Pi_{\mathcal{F}}$ is the $L^2(\mu^\star)$-projection onto $\mathcal{F}$. Since this projection is nonexpansive in the norm where the linearized Bellman update contracts, the projected operator can inherit local contraction. We make this precise in Section 3.

The ideal update (2) is not directly implementable, because $\mu^\star$ depends on the unknown optimal policy. Instead, it defines the population norm that the fitted regression should approximate. Section 4 implements this idea by estimating Bellman targets from behavior data and weighting observations by density ratios for the stationary distribution of the current softmax policy $\pi_{Q^{(k)}}$. Near $Q^\star$, the stationary distribution of $\pi_{Q^{(k)}}$ is close to $\mu^\star$, so iterative reweighting approximates the ideal stationary projection. Section 5 gives finite-sample theory.

3 Local Contraction and Convergence of Soft Q-Iteration

3.1 Local approximation and contraction

We first control the linearization remainder in (1) through a local smoothness condition around $Q^\star$. This shows that, near $Q^\star$, the projected soft Bellman operator $\mathcal{T}_{\mathcal{F}}$ is a locally contractive perturbation of the policy-evaluation operator $\mathcal{T}^{\text{eval}}_{Q^\star}$.

For $r > 0$ and $Q_0 \in L^2(\mu^\star)$, write

$$\mathbb{B}_{\mathcal{F}}(r, Q_0) := \{f \in \mathcal{F} : \|f - Q_0\|_{2,\mu^\star} \le r\}, \qquad \mathcal{H}^\star_{\mathcal{F}} := \{Q_1 - Q_2 : Q_1, Q_2 \in \mathcal{F} \cup \{Q^\star\}\}.$$

For $R > 0$, define the $Q^\star$-star hull of the local model ball

$$\mathbb{S}^\star_{\mathcal{F}}(R) := \{Q^\star + t(f - Q^\star) : f \in \mathbb{B}_{\mathcal{F}}(R, Q^\star),\ t \in [0,1]\},$$

and define the local Bellman derivative modulus

$$\omega_{\mathcal{T}}(R) := \sup_{\substack{Q\in\mathbb{S}^\star_{\mathcal{F}}(R)\\ H\in\mathcal{H}^\star_{\mathcal{F}},\ H\neq 0}} \frac{\|(D\mathcal{T}(Q) - D\mathcal{T}(Q^\star))[H]\|_{2,\mu^\star}}{\|H\|_{2,\mu^\star}}.$$

C1) Stationary action positivity. Exists $\pi_{\min} > 0$ such that $\pi^\star(A \mid S) \ge \pi_{\min}$ $\mu^\star$-almost surely.

C2) Model class regularity. $\mathcal{F} \subseteq L^2(\mu^\star)$ is convex, closed, and uniformly bounded.

C3) Local Bellman curvature. $\omega_{\mathcal{T}}(R) \to 0$ as $R \downarrow 0$.

Condition C1 gives stationary action positivity, and Condition C2 is standard for convex linear-model constraints, such as ℓ^1- and ℓ^2-balls. Condition C3 is the only local smoothness requirement: it asks that the derivative of the soft Bellman operator be continuous, in stationary norm, along directions relevant to the local projected iteration. Appendices D.3 and H.2 give sufficient conditions under which $\omega_{\mathcal{T}}(R) \lesssim R^\alpha$ for some $\alpha \in (0,1]$, using L^∞–L^2 interpolation for finite-dimensional, Hölder, Sobolev, and RKHS classes. Appendix H.4.1 shows that, under action-gap conditions, the hard-max limit can also be handled: the local curvature modulus remains small on shrinking neighborhoods as $\tau \downarrow 0$.

Define the local contraction modulus and radius by

$$\rho_{\text{loc}}(R) := \gamma + \omega_{\mathcal{T}}(R), \qquad r_{\text{loc}} := \sup\{R > 0 : \rho_{\text{loc}}(R) < 1\}.$$

Lemma 1 (Linearization remainder). *Under Conditions C1–C3, for any $R < r_{\text{loc}}$ and any $Q \in \mathbb{B}_{\mathcal{F}}(R, Q^\star)$,*

$$\left\|\mathcal{T}(Q) - \mathcal{T}^{\text{eval}}_{Q^\star}(Q)\right\|_{2,\mu^\star} \le \omega_{\mathcal{T}}(R)\,\|Q - Q^\star\|_{2,\mu^\star}.$$

Combining the preceding lemma with the global γ-contraction of $\mathcal{T}^{\mathrm{eval}}_{Q^\star}$ in $L^2(\mu^\star)$ gives a local contraction bound for $\mathcal{T}_{\mathcal{F}}$. By definition of r_{loc}, the linearization error is smaller than the contraction margin on every ball of radius $R < r_{\mathrm{loc}}$. Hence the contraction argument applies on $\mathbb{B}_{\mathcal{F}}(r_{\mathrm{loc}}, Q^\star)$, provided this set is nonempty, namely when

$$\mathrm{dist}(\mathcal{F}, Q^\star) := \|\Pi_{\mathcal{F}} Q^\star - Q^\star\|_{2,\mu^\star} \le r_{\mathrm{loc}}.$$

Theorem 1 (Local contraction of the projected soft Bellman operator)**.** *Assume Conditions C1–C3 hold. For any $R \in [\mathrm{dist}(\mathcal{F}, Q^\star),\, r_{\mathrm{loc}})$, $\mathcal{T}_{\mathcal{F}} = \Pi_{\mathcal{F}}\mathcal{T}$ is a $\rho_{\mathrm{loc}}(R)$-contraction on $\mathbb{B}_{\mathcal{F}}(R, Q^\star)$: for all $Q_1, Q_2 \in \mathbb{B}_{\mathcal{F}}(R, Q^\star)$,*

$$\|\mathcal{T}_{\mathcal{F}}(Q_1) - \mathcal{T}_{\mathcal{F}}(Q_2)\|_{2,\mu^\star} \;\le\; \rho_{\mathrm{loc}}(R)\, \|Q_1 - Q_2\|_{2,\mu^\star}.$$

3.2 Local convergence of soft Q-iteration

If the approximation error is small relative to the local contraction margin, Banach's fixed-point theorem gives a unique projected fixed point near $Q^\star$. We impose the following condition.

C4) Approximate realizability. There exists $R_{\mathrm{fp}} \in (0, r_{\mathrm{loc}})$ such that

$$\mathrm{dist}(\mathcal{F}, Q^\star) \le \{1 - \rho_{\mathrm{loc}}(R_{\mathrm{fp}})\} R_{\mathrm{fp}}.$$

By Lemma 9, Condition C4 ensures that $\mathcal{T}_{\mathcal{F}}$ has a fixed point $Q^\dagger \in \mathcal{F}$ inside the local contraction region:

$$Q^\dagger = \mathcal{T}_{\mathcal{F}}(Q^\dagger), \qquad \varepsilon_{\mathcal{F}} := \|Q^\dagger - Q^\star\|_{2,\mu^\star}.$$

We next show that soft Q-iteration converges linearly when initialized near $Q^\dagger$. Since the contraction region is centered at $Q^\star$, any ball $\mathbb{B}_{\mathcal{F}}(r, Q^\dagger)$ with $r + \varepsilon_{\mathcal{F}} < r_{\mathrm{loc}}$ is contained in the local contraction region around $Q^\star$.

Theorem 2 (Local linear convergence around the projected soft optimum)**.** *Assume Conditions C1–C4 hold. Let $r > 0$ satisfy $r + \varepsilon_{\mathcal{F}} < r_{\mathrm{loc}}$. Then, for any initialization $Q^{(0)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\dagger)$ and all $k \ge 0$,*

$$\|Q^{(k)} - Q^\dagger\|_{2,\mu^\star} \le \{\rho_{\mathrm{loc}}(r + \varepsilon_{\mathcal{F}})\}^k \|Q^{(0)} - Q^\dagger\|_{2,\mu^\star}.$$

The theorem shows that soft Q-iteration is locally contractive around $Q^\dagger$: within any admissible ball $\mathbb{B}_{\mathcal{F}}(r, Q^\dagger)$ with $r + \varepsilon_{\mathcal{F}} < r_{\mathrm{loc}}$, the iterates move toward $Q^\dagger$ at rate $\rho_{\mathrm{loc}}(r + \varepsilon_{\mathcal{F}})$. As the iterates approach $Q^\dagger$, they enter smaller basins with sharper contraction moduli. In the realizable case, the modulus approaches γ at a geometric rate; see Corollary 2 in Appendix E.4.

4 Stationary-Reweighted Soft FQI

The preceding section identifies the ideal stationary-norm update $\mathcal{T}_{\mathcal{F}} = \Pi_{\mathcal{F}}\mathcal{T}$. Stationary-reweighted soft FQI approximates this update from behavior data by weighting each Bellman regression with an estimated stationary density ratio. This section gives the finite-sample algorithm.

Algorithm. Given offline transitions $\mathcal{D}_n := \{(S_i, A_i, R_i, S_i')\}_{i=1}^n$ sampled from ν_b and a function class $\mathcal{F}$, initialize $\widehat{Q}^{(0)}$. For the induced policy $\pi_{\widehat{Q}^{(0)}}$, estimate weights $\widehat{d}^{(0)}$ targeting the stationary ratio $\mathrm{d}\mu_{\pi_{\widehat{Q}^{(0)}}}/\mathrm{d}\nu_b$.

For each $k \in \{0, \ldots, K-1\}$, form the empirical soft Bellman targets

$$\widehat{y}_i^{(k)} = R_i + \gamma\tau \log \sum_{a'} \exp\big(\widehat{Q}^{(k)}(S_i', a')/\tau\big).$$

Then update the Q-estimate by weighted least squares:

$$\widehat{Q}^{(k+1)} \in \operatorname*{arg\,min}_{Q \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \widehat{d}^{(k)}(S_i, A_i) \left(\widehat{y}_i^{(k)} - Q(S_i, A_i)\right)^2.$$

Finally, estimate weights $\widehat{d}^{(k+1)}$ for the updated policy $\pi_{\widehat{Q}^{(k+1)}}$ and continue.

Computation. Each iteration adds a density-ratio estimation step to the usual Bellman regression step. In practice, neither the Q-function nor the density ratio must be solved to completion at every iteration. One can instead parametrize both Q and d with neural networks and alternate between one or a few stochastic-gradient updates for each, as in deep fitted Q-learning or boosting-style fitted value iteration [59, 64].

Estimating stationary ratios. The stationary state-action density ratio can be estimated using existing off-policy evaluation tools, including DICE-style saddle-point methods [41, 49, 50, 88, 39, 40], minimax weight estimators [67, 68], and projected or balancing-weight methods [79]; see Appendix H.3. With linear or RKHS critics, these objectives often reduce to tractable linear-system or closed-form updates [17, 67, 79, 53]. Existing finite-sample guarantees apply under conditions such as realizability, completeness, coverage, and identification [68]. Stationary ratios are the undiscounted analogues of discounted occupancy ratios, which are central to importance-sampling, doubly robust, and efficient policy-value estimators [31, 63, 34, 35, 5]. In our setting, the ratio plays a different role: it stabilizes the Bellman regression geometry rather than only debiasing a final policy-value estimate.

Initialization, warm starts, and temperature annealing. Our guarantees are local: they apply once the iterates lie in a basin where the stationary projected Bellman update is contractive. Standard soft FQI can be used as a warm start for reaching this basin. For example, under approximate Bellman completeness, standard soft FQI may approach $Q^\dagger$ up to an inherent Bellman error [48]; if this error is smaller than the contraction radius, the local guarantee applies from that point onward. After the switch to stationary weighting, the recursion contracts toward the stationary projected fixed point, and the finite-sample bound no longer accumulates the classical inherent Bellman-error term.

Temperature annealing provides another basin-entry heuristic, inspired by classical continuation and deterministic annealing methods [3, 60]. At large temperature τ, the soft Bellman update is smoother, the local contraction basin can be larger, and the optimal soft policy is closer to uniform. Thus, one can start from a simple initialization, approximate the initial stationary ratio by that of the uniform policy, and then gradually decrease τ, warm-starting each stage from the previous solution while updating the stationary-ratio estimate. If consecutive basins overlap, the iterates remain locally stable along the annealing path. Appendix H.4 gives a population-level continuation argument and shows that, under an action-gap condition, this path can approach the hard-max limit. A full finite-sample analysis of the annealing scheme is left for future work.

5 Finite-Sample Convergence Analysis

We prove the finite-sample guarantee for stationary-reweighted soft FQI. Theorem 3 shows that, within an admissible local basin, the iterates contract toward $Q^\dagger$ up to statistical error from empirical Bellman regression and geometrically damped weight-estimation error. The proof combines: (i) an inexact local Picard recursion, (ii) a high-probability bound for one weighted Bellman regression, and (iii) a localization argument that keeps the iterates inside the contraction basin.

5.1 Inexact local Picard iteration

We first establish a local analogue of the classical FQI recursion of Munos and Szepesvári [48]. Since contraction holds only inside a local basin, the per-iteration errors need not be zero, but must be small relative to the contraction margin.

Lemma 2 (Local inexact iteration error bound)**.** *Assume Conditions C1–C4. Fix $r > 0$ satisfying $r + \varepsilon_{\mathcal{F}} < r_{\mathrm{loc}}$, and set $\rho_r := \rho_{\mathrm{loc}}(r + \varepsilon_{\mathcal{F}})$. Suppose that, for some nonnegative sequence $\{\xi_k\}_{k\geq 0}$, the inexact iterates satisfy*

$$\|\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\|_{2,\mu^\star} \leq \xi_k, \quad \|\widehat{Q}^{(0)} - Q^\dagger\|_{2,\mu^\star} \leq r, \quad \xi_k \leq (1-\rho_r)r.$$

Then, for every $k \geq 0$,

$$\|\widehat{Q}^{(k)} - Q^\dagger\|_{2,\mu^\star} \leq \rho_r^k \|\widehat{Q}^{(0)} - Q^\dagger\|_{2,\mu^\star} + \sum_{j=0}^{k-1} \rho_r^{k-1-j}\xi_j.$$

The condition $\xi_k \leq (1-\rho_r)r$ is the local-basin slack condition: at the boundary of the basin, contraction pulls the iterate inward by $(1-\rho_r)r$, and the one-step error must fit within this margin. Under this condition, Lemma 2 yields linear convergence up to the inexactness floor $\sum_{j=0}^{k-1} \rho_r^{k-1-j}\xi_j$.

5.2 Finite-sample bounds on the per-iteration error

We next bound the per-iteration inexactness $\|\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\|_{2,\mu^\star}$ with high probability, separating sampling noise, approximation error, and weight-estimation error. We express the ERM rates through metric entropy. Let $N(\varepsilon, \mathcal{F}, L^2(P))$ denote the $L^2(P)$-covering number, and define

$$\mathcal{J}(\delta, \mathcal{F}) := \int_0^\delta \sup_P \sqrt{\log N(\varepsilon, \mathcal{F}, L^2(P))}\, d\varepsilon, \qquad \delta_n := \inf\left\{\delta > 0 : \mathcal{J}(\delta, \mathcal{F}) \le \sqrt{n}\,\delta^2\right\}.$$

Then δ_n yields the standard ERM rates over $\mathcal{F}$ [78]; for example, $\delta_n \asymp \sqrt{s\log(d/s)/n}$ for s-sparse linear models, $\delta_n \asymp \sqrt{V\log(n/V)/n}$ for VC-subgraph classes, and $\delta_n \asymp n^{-s/(2s+d)}$ for Hölder/Sobolev classes of smoothness s in dimension $d \le 2s$ [78].

C5) Bounded regression objects. There exists $M < \infty$ such that $|R| \le M$ and $\sup_{f\in\mathcal{F}} \|f\|_\infty \le M$.

C6) Coverage and bounded weights. $\mu^\star \ll \nu_b$ and there exists $1 \le \kappa_{\mathrm{cov}} < \infty$ such that, for all $0 \le k \le K-1$,

$$\|d^\star\|_\infty \le \kappa_{\mathrm{cov}}, \qquad \|\widehat{d}^{(k)}\|_\infty \le \kappa_{\mathrm{cov}}, \qquad \|(\widehat{d}^{(k)} - d^\star)/\sqrt{d^\star}\|_\infty \le \kappa_{\mathrm{cov}}.$$

C7) Sample splitting. The weight estimators $\{\widehat{d}^{(k)}\}_{k=0}^K$ are computed using data independent of the batch $\mathcal{D}_n$ used for the regression updates.

C8) Entropy integral regularity. $\mathcal{J}(\infty, \mathcal{F}) < \infty$, $\mathcal{J}(\delta, \mathcal{F})/\{\delta\sqrt{\mathrm{loglog}(1/\delta)}\} \to \infty$ as $\delta \to 0$.

Condition C5 simplifies the analysis and can be relaxed using truncation and tail conditions [78]. Condition C6 imposes bounded target–behavior coverage and keeps the estimated weights controlled. It is the analogue of coverage or concentrability assumptions in offline RL [83, 67]. Condition C7 ensures independence between weight estimation and the regression sample and can be removed by cross-fitting [52, 22, 71]. It is only a split between the data used to estimate the weights and the data used for the Bellman regressions; the FQI regression iterates may all be fitted on the same $\mathcal{D}_n$, since the empirical-process event below is uniform over the regression and target function classes. Finally, Condition C8 controls localized Rademacher complexities, following Munos and Szepesvári [48], and holds for classes with polynomial metric entropy.

Fix $K \ge 1$ and $\eta \in (0,1)$, and define the statistical regression error

$$\delta_{\mathrm{stat}}(n, \eta, K) := \delta_n + \sqrt{\frac{\log(eK/\eta)}{n}}, \qquad \mathcal{H}_{\mathcal{F}} := \{f - f' : f, f' \in \mathcal{F}\}.$$

For each iteration, define the relative distortion of the weighted quadratic norm by

$$\chi_{\mathcal{H},k} := \sup_{\substack{h\in\mathcal{H}_{\mathcal{F}} \\ E_{\nu_b}[d^\star h^2] > 0}} \left| \frac{E_{\nu_b}[(\widehat{d}^{(k)} - d^\star)h^2]}{E_{\nu_b}[d^\star h^2]} \right|,$$

and the residual-interaction term

$$\omega_{\mathrm{Bell},d^\star}(k) := \left\| \frac{\widehat{d}^{(k)} - d^\star}{\sqrt{d^\star}} \{\mathcal{T}\widehat{Q}^{(k)} - \mathcal{T}_{\mathcal{F}}\widehat{Q}^{(k)}\} \right\|_{2,\nu_b}.$$

C9) Weighted-loss curvature stability. Across the K iterations, $\bar{\chi}_{\mathcal{H}} := \max_{0\le k\le K-1} \chi_{\mathcal{H},k} < 1$.

Condition C9 ensures that, for all $h \in \mathcal{H}_{\mathcal{F}}$ and $0 \le k \le K-1$,

$$E_{\nu_b}[\widehat{d}^{(k)} h^2] \ge (1 - \chi_{\mathcal{H},k}) E_{\nu_b}[d^\star h^2].$$

Thus the fitted weighted loss preserves the ideal stationary curvature up to the relative factor $1 - \chi_{\mathcal{H},k}$ on fitted directions. A simple sufficient condition is $\max_{0\le k\le K-1} \|\widehat{d}^{(k)}/d^\star - 1\|_\infty < 1$, which holds with probability tending to one, with $\bar{\chi}_{\mathcal{H}} = o_p(1)$, if the weights are uniformly consistent. More refined L^2-relative conditions are given in Appendix H.2.

Lemma 3 (Statistical accuracy of one weighted regression)**.** *Assume Conditions C5–C9. There exists $C = C(M, \tau \log |\mathcal{A}|) < \infty$ such that, for all $\eta \in (0,1)$, with probability at least $1-\eta$, simultaneously for all $0 \le k \le K-1$,*

$$\left\|\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\right\|_{2,\mu^\star} \le \frac{C}{1-\chi_{\mathcal{H},k}}\Big[\kappa_{\mathrm{cov}}\delta_{\mathrm{stat}}(n,\eta,K) + \omega_{\mathrm{Bell},d^\star}(k)\Big].$$

The first term is the usual ERM error over $\mathcal{F}$. Estimated weights enter in two ways. The factor $(1-\chi_{\mathcal{H},k})^{-1}$ is the curvature loss from using the estimated weighted quadratic form; it is close to one under uniform weight consistency, but can be large if weights distort curvature on fitted directions. The term $\omega_{\mathrm{Bell},d^\star}(k)$ captures the interaction between weight error and the Bellman approximation residual. For $\varepsilon_{\mathrm{Bell}} := \sup_{Q\in\mathcal{F}} \|\mathcal{T}Q - \mathcal{T}_{\mathcal{F}}Q\|_{2,\mu^\star}$ and $\varepsilon_{\mathrm{Bell},\infty} := \sup_{Q\in\mathcal{F}} \|\mathcal{T}Q - \mathcal{T}_{\mathcal{F}}Q\|_{\infty}$,

$$\omega_{\mathrm{Bell},d^\star}(k) \le \min\left\{\left\|\frac{\widehat{d}^{(k)}}{d^\star} - 1\right\|_{\infty}\varepsilon_{\mathrm{Bell}},\ \left\|\frac{\widehat{d}^{(k)}}{d^\star} - 1\right\|_{2,\mu^\star}\varepsilon_{\mathrm{Bell},\infty}\right\}.$$

Thus the residual-interaction term is small when the weights are accurate or when $\mathcal{F}$ has small inherent Bellman error [48].

5.3 Finite-sample convergence bound

Combining Lemma 2 with the per-iteration bound in Lemma 3 gives the main finite-sample result. Because the contraction argument is local, we work on a radius where the population contraction dominates the one-step regression errors. For $r > 0$, let $C = C(M, \tau \log |\mathcal{A}|)$ and define

$$\rho_K(r) := \rho_{\mathrm{loc}}(r + \varepsilon_{\mathcal{F}}), \qquad \bar{\omega}_{\mathrm{Bell},d^\star} := \max_{0\le j\le K-1} \omega_{\mathrm{Bell},d^\star}(j).$$

The worst-case one-step regression error is

$$\bar{\xi}_K := \frac{C}{1-\bar{\chi}_{\mathcal{H}}}\left\{\kappa_{\mathrm{cov}}\delta_{\mathrm{stat}}(n,\eta,K) + \bar{\omega}_{\mathrm{Bell},d^\star}\right\}.$$

C10) Admissible localization radius. There exists $r > 0$ such that

$$r + \varepsilon_{\mathcal{F}} < r_{\mathrm{loc}}, \qquad \rho_K(r) < 1, \qquad \bar{\xi}_K \le (1-\rho_K(r))r.$$

Condition C10 is a finite-sample localization requirement: the one-step perturbation must be smaller than the stability margin of some local contraction ball. In particular, if the statistical and weight-estimation errors vanish, then for any fixed $r > 0$ satisfying $r + \varepsilon_{\mathcal{F}} < r_{\mathrm{loc}}$ and $\rho_K(r) < 1$, the slack condition $\bar{\xi}_K \le (1-\rho_K(r))r$ holds for all sufficiently large n.

Our main result is stated below. For a fixed admissible radius r, write $\rho_K := \rho_K(r)$, and define the geometrically averaged Bellman-residual interaction

$$\Omega_{\mathrm{Bell},d^\star}(k) := (1-\rho_K)\sum_{j=0}^{k-1} \rho_K^{k-1-j}\omega_{\mathrm{Bell},d^\star}(j).$$

Theorem 3 (Local convergence with geometrically damped weight error)**.** *Assume Conditions C1–C10 hold. Fix any admissible $r > 0$, use the corresponding $\rho_K = \rho_K(r)$, and assume $\widehat{Q}^{(0)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\dagger)$. Then, with probability at least $1-\eta$, for all $1 \le k \le K$,*

$$\left\|\widehat{Q}^{(k)} - Q^\dagger\right\|_{2,\mu^\star} \le \rho_K^k\left\|\widehat{Q}^{(0)} - Q^\dagger\right\|_{2,\mu^\star} + \frac{C}{(1-\rho_K)(1-\bar{\chi}_{\mathcal{H}})}\Big\{\kappa_{\mathrm{cov}}\delta_{\mathrm{stat}}(n,\eta,K) + \Omega_{\mathrm{Bell},d^\star}(k)\Big\}.$$

As a useful special case, ordinary soft FQI is locally stable under on-policy stationary sampling.

Corollary 1 (On-policy stationary sampling)**.** *Assume the conditions of Theorem 3 hold, with $\nu_b = \mu^\star$ and unit weights $\hat{d}^j \equiv d^\star \equiv 1$. Then $\chi_{\mathcal{H},j} = 0$ and $\omega_{\mathrm{Bell},d^\star}(j) = 0$ for all j, and with probability at least $1-\eta$, for all $1 \le k \le K$,*

$$\left\|\widehat{Q}^{(k)} - Q^\dagger\right\|_{2,\mu^\star} \le \rho_K^k\left\|\widehat{Q}^{(0)} - Q^\dagger\right\|_{2,\mu^\star} + \frac{C}{1-\rho_K}\,\delta_{\mathrm{stat}}(n,\eta,K).$$

Thus, under on-policy stationary sampling, unweighted soft FQI converges locally linearly up to statistical error. The remaining terms in Theorem 3 quantify the cost of approximating this stationary projection off policy.

Structure of the bound. Theorem 3 is an inexact Picard recursion in the locally contractive stationary norm. The initialization error contracts at rate ρ_K. The statistical error and Bellman-residual interaction enter additively with the usual stability factor $(1-\rho_K)^{-1}$ and the regression-curvature loss $(1-\bar{\chi}_{\mathcal{H}})^{-1}$. Unlike classical FQI bounds based on Bellman completeness, the result uses stationary ratio weighting and approximate realizability. Appendix H.3 reviews existing methods for estimating stationary ratios, including DICE-style, minimax, and balancing-weight approaches, and summarizes representative finite-sample guarantees [67, 68].

The term $\Omega_{\mathrm{Bell},d^\star}(k)$ tracks how weight error interacts with Bellman approximation error over the k iterations. It is a geometric average of the per-step terms $\omega_{\mathrm{Bell},d^\star}(j)$: errors from recent iterations receive more weight, while earlier errors are damped by the contraction factor ρ_K. Thus imperfect weights are most harmful when they align with the Bellman residual $\mathcal{T}\widehat{Q}^{(j)} - \mathcal{T}_{\mathcal{F}}\widehat{Q}^{(j)}$. If these interactions decay geometrically, then $\Omega_{\mathrm{Bell},d^\star}(k)$ decays at the slower of that decay rate and ρ_K. Appendix G.3 gives a companion bound for the coupled case $\omega_{\mathrm{Bell},d^\star}(k) \lesssim \|\widehat{Q}^{(k)} - Q^\dagger\|_{2,\mu^\star}^{\kappa} + \delta_{\mathrm{wt}}$.

6 Experimental investigation

We use two controlled simulations to illustrate the mechanism predicted by the local theory. The Garnet experiment in Appendix B.1 isolates norm mismatch and temperature annealing in a finite MDP. The main continuous offline RL experiment adds nonlinear state dynamics, controlled behavior–target shift, estimated stationary ratios, and finite-sample weight stabilization.

Continuous offline RL simulation. We evaluate a continuous-state navigation MDP designed to separate Bellman incompleteness from off-policy distribution shift. Across ten behavior regimes, from nearly on-policy to severe shift, we compare unweighted soft FQI, stabilized oracle stationary weighting, and estimated local RBF-polynomial stationary weighting using a deliberately Bellman-incomplete linear state-action Q-class. A grid solve at $\tau = 10^{-3}$ provides $Q_\tau^\star$, the target stationary distribution, and oracle density ratios. The estimated weights use a DICE-style moment estimator with finite-sample stabilization. We report direct low-temperature fitting at $\tau = 10^{-3}$; Appendix C gives environment, weighting, and annealing details.

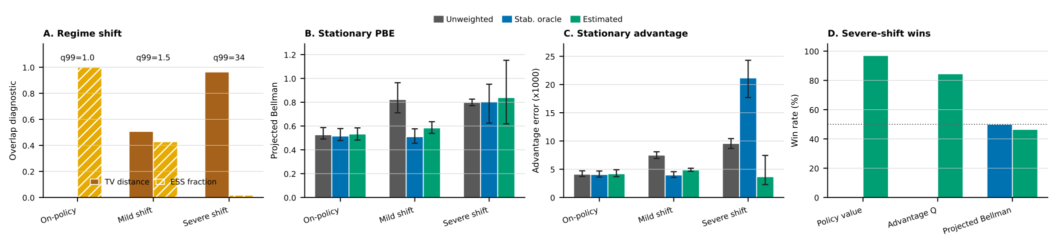

Figure 1: Continuous offline RL. (A) Overlap diagnostics over ten behavior regimes from near-on-policy to severe off-policy shift. (B) Stationary projected Bellman error. (C) Advantage-centered stationary Q-error. (D) Severe-shift win rates against unweighted soft FQI; unweighted is the reference. Points and intervals in (B)–(C) show medians and IQRs over 500 datasets.

Figure 1 reports stationary projected Bellman error, $\|Q - \mathcal{T}_{\mathcal{F}}(Q)\|_{2,\mu^\star}$, and advantage-centered stationary Q-RMSE, $\|(Q - Q_\tau^\star) - \pi^\star(Q - Q_\tau^\star)\|_{2,\mu^\star}$, which removes within-state shifts and captures action-relevant error. Weighting is neutral near on-policy but reduces target-stationary Bellman and advantage errors as behavior–target shift increases. Under severe shift, poor coverage makes exact ratios highly variable, so stabilized approximate weighting can improve action-relevant error even when it does not uniformly reduce projected Bellman error. Estimated local RBF-polynomial weights beat unweighted FQI on advantage error in 407 of 500 seeds and policy value in 488 of 500 seeds, but on projected Bellman residual in only 234 of 500 seeds. Expanded discussion is in Appendix B.2.

Comparison with minimax Bellman-residual fitting. Appendix B.3 compares stationary weighting with a regularized minimax Bellman-residual baseline using the same linear Q-features and the same RBF-polynomial critic dictionary used for ratio estimation. The methods show complementary strengths: estimated stationary weights track the stabilized oracle correction when coverage is adequate, while minimax residual fitting is competitive in some severe-shift regimes where ratio estimation is difficult. We evaluate minimax by Bellman residual, advantage error, and true-Q error.

Takeaway. The results suggest that Bellman completeness is not the only route to stable soft fitted value iteration: aligning the projection norm with the local stationary geometry can recover contraction under misspecification. The guarantees remain local and finite-action. In particular, constants depend on minimum action probabilities, the present proof does not directly cover continuous actions, and without an action gap the low-temperature contraction radius can shrink. High-dimensional offline RL benchmarks are left for future work.

References

[1] Robert A Adams and John JF Fournier. *Sobolev spaces*, volume 140. Elsevier, 2003.

[2] Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron C Courville, and Marc Bellemare. Deep reinforcement learning at the edge of the statistical precipice. *Advances in neural information processing systems*, 34:29304–29320, 2021.

[3] Eugene L Allgower and Kurt Georg. Numerical continuation methods: an introduction, 1990.

[4] Philip Amortila, Nan Jiang, and Tengyang Xie. A variant of the wang-foster-kakade lower bound for the discounted setting. *arXiv preprint arXiv:2011.01075*, 2020.

[5] Philip Amortila, Dylan J Foster, Nan Jiang, Ayush Sekhari, and Tengyang Xie. Harnessing density ratios for online reinforcement learning. *arXiv preprint arXiv:2401.09681*, 2024.

[6] Leemon Baird et al. Residual algorithms: Reinforcement learning with function approximation. In *Proceedings of the twelfth international conference on machine learning*, pages 30–37, 1995.

[7] Heinz H Bauschke and Patrick L Combettes. *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*. CMS Books in Mathematics / Ouvrages de mathématiques de la SMC. Springer, New York, NY, 2011.

[8] Richard Bellman. Dynamic programming. *science*, 153(3731):34–37, 1966.

[9] Andrew Bennett, Nathan Kallus, Xiaojie Mao, Whitney Newey, Vasilis Syrgkanis, and Masatoshi Uehara. Minimax instrumental variable regression and l_2 convergence guarantees without identification or closedness. In *The Thirty Sixth Annual Conference on Learning Theory*, pages 2291–2318. PMLR, 2023.

[10] Dimitri Bertsekas and John N Tsitsiklis. *Neuro-Dynamic Programming*. Athena Scientific, 1996.

[11] Shalabh Bhatnagar, Richard S Sutton, Mohammad Ghavamzadeh, and Mark Lee. Natural actor–critic algorithms. *Automatica*, 45(11):2471–2482, 2009.

[12] Aurelien Bibaut, Maya Petersen, Nikos Vlassis, Maria Dimakopoulou, and Mark van der Laan. Sequential causal inference in a single world of connected units. *arXiv preprint arXiv:2101.07380*, 2021.

[13] Jonathan Chang, Kaiwen Wang, Nathan Kallus, and Wen Sun. Learning bellman complete representations for offline policy evaluation. In *International Conference on Machine Learning*, pages 2938–2971. PMLR, 2022.

[14] Jinglin Chen and Nan Jiang. Information-theoretic considerations in batch reinforcement learning. In *International conference on machine learning*, pages 1042–1051. PMLR, 2019.

[15] Bo Dai, Albert Shaw, Lihong Li, Lin Xiao, Niao He, Zhen Liu, Jianshu Chen, and Le Song. Sbeed: Convergent reinforcement learning with nonlinear function approximation. In *International conference on machine learning*, pages 1125–1134. PMLR, 2018.

[16] Kristopher De Asis, Eric Graves, and Richard S Sutton. Value-aware importance weighting for off-policy reinforcement learning. In *Conference on Lifelong Learning Agents*, pages 745–763. PMLR, 2023.

[17] Nishanth Dikkala, Greg Lewis, Lester Mackey, and Vasilis Syrgkanis. Minimax estimation of conditional moment models. *Advances in Neural Information Processing Systems*, 33: 12248–12262, 2020.

[18] Yaqi Duan, Zeyu Jia, and Mengdi Wang. Minimax-optimal off-policy evaluation with linear function approximation. In *International Conference on Machine Learning*, pages 2701–2709. PMLR, 2020.

[19] Damien Ernst, Pierre Geurts, and Louis Wehenkel. Tree-based batch mode reinforcement learning. *Journal of Machine Learning Research*, 6, 2005.

[20] Jianqing Fan, Zhaoran Wang, Yuchen Xie, and Zhuoran Yang. A theoretical analysis of deep q-learning. In *Learning for dynamics and control*, pages 486–489. PMLR, 2020.

[21] Amir-massoud Farahmand, Csaba Szepesvári, and Rémi Munos. Error propagation for approximate policy and value iteration. *Advances in neural information processing systems*, 23, 2010.

[22] Dylan J Foster and Vasilis Syrgkanis. Orthogonal statistical learning. *The Annals of Statistics*, 51(3):879–908, 2023.

[23] Dylan J Foster, Akshay Krishnamurthy, David Simchi-Levi, and Yunzong Xu. Offline reinforcement learning: Fundamental barriers for value function approximation. *arXiv preprint arXiv:2111.10919*, 2021.

[24] Scott Fujimoto, David Meger, and Doina Precup. Off-policy deep reinforcement learning without exploration. In *International conference on machine learning*, pages 2052–2062. PMLR, 2019.

[25] Matthieu Geist, Bruno Scherrer, and Olivier Pietquin. A theory of regularized markov decision processes. In *International conference on machine learning*, pages 2160–2169. PMLR, 2019.

[26] Geoffrey J Gordon. Stable function approximation in dynamic programming. In *Machine learning proceedings 1995*, pages 261–268. Elsevier, 1995.

[27] Tuomas Haarnoja, Haoran Tang, Pieter Abbeel, and Sergey Levine. Reinforcement learning with deep energy-based policies. In *International conference on machine learning*, pages 1352–1361. PMLR, 2017.

[28] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International conference on machine learning*, pages 1861–1870. Pmlr, 2018.

[29] Assaf Hallak and Shie Mannor. Consistent on-line off-policy evaluation. In *International Conference on Machine Learning*, pages 1372–1383. PMLR, 2017.

[30] Assaf Hallak, Aviv Tamar, Rémi Munos, and Shie Mannor. Generalized emphatic temporal difference learning: Bias-variance analysis. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 30, 2016.

[31] Nan Jiang and Lihong Li. Doubly robust off-policy value evaluation for reinforcement learning. In *International conference on machine learning*, pages 652–661. PMLR, 2016.

[32] Chi Jin, Qinghua Liu, and Sobhan Miryoosefi. Bellman eluder dimension: New rich classes of RL problems, and sample-efficient algorithms. *Advances in neural information processing systems*, 34:13406–13418, 2021.

[33] Chi Jin, Zhuoran Yang, Zhaoran Wang, and Michael I Jordan. Provably efficient reinforcement learning with linear function approximation. *Mathematics of Operations Research*, 48(3): 1496–1521, 2023.

[34] Nathan Kallus and Masatoshi Uehara. Double reinforcement learning for efficient off-policy evaluation in markov decision processes. *Journal of Machine Learning Research*, 21(167):1–63, 2020.

[35] Nathan Kallus and Masatoshi Uehara. Efficiently breaking the curse of horizon in off-policy evaluation with double reinforcement learning. *Operations Research*, 70(6):3282–3302, 2022.

[36] Guanghui Lan. Policy mirror descent for reinforcement learning: Linear convergence, new sampling complexity, and generalized problem classes. *Mathematical programming*, 198(1): 1059–1106, 2023.

[37] Alessandro Lazaric, Mohammad Ghavamzadeh, and Rémi Munos. Finite-sample analysis of least-squares policy iteration. *The Journal of Machine Learning Research*, 13(1):3041–3074, 2012.

[38] Hoang Le, Cameron Voloshin, and Yisong Yue. Batch policy learning under constraints. In *International Conference on Machine Learning*, pages 3703–3712. PMLR, 2019.

[39] Jongmin Lee, Wonseok Jeon, Byungjun Lee, Joelle Pineau, and Kee-Eung Kim. Optidice: Offline policy optimization via stationary distribution correction estimation. In *International Conference on Machine Learning*, pages 6120–6130. PMLR, 2021.

[40] Jongmin Lee, Cosmin Paduraru, Daniel J Mankowitz, Nicolas Heess, Doina Precup, Kee-Eung Kim, and Arthur Guez. Coptidice: Offline constrained reinforcement learning via stationary distribution correction estimation. *arXiv preprint arXiv:2204.08957*, 2022.

[41] Qiang Liu, Lihong Li, Ziyang Tang, and Dengyong Zhou. Breaking the curse of horizon: Infinite-horizon off-policy estimation. *Advances in neural information processing systems*, 31, 2018.

[42] A Rupam Mahmood, Huizhen Yu, Martha White, and Richard S Sutton. Emphatic temporal-difference learning. *arXiv preprint arXiv:1507.01569*, 2015.

[43] Mohammad Mehrabi and Stefan Wager. Off-policy evaluation in markov decision processes under weak distributional overlap. *arXiv preprint arXiv:2402.08201*, 2024.

[44] Francisco S Melo and M Isabel Ribeiro. Convergence of q-learning with linear function approximation. In *2007 European control conference (ECC)*, pages 2671–2678. IEEE, 2007.

[45] Shahar Mendelson and Joseph Neeman. Regularization in kernel learning. *The Annals of Statistics*, 38(1):526–565, 2010.

[46] Sean Meyn. The projected bellman equation in reinforcement learning. *IEEE Transactions on Automatic Control*, 69(12):8323–8337, 2024.

[47] Rémi Munos. Error bounds for approximate value iteration. In *Proceedings of the National Conference on Artificial Intelligence*, volume 20, page 1006. Menlo Park, CA; Cambridge, MA; London; AAAI Press; MIT Press; 1999, 2005.

[48] Rémi Munos and Csaba Szepesvári. Finite-time bounds for fitted value iteration. *Journal of Machine Learning Research*, 9(5), 2008.

[49] Ofir Nachum, Yinlam Chow, Bo Dai, and Lihong Li. Dualdice: Behavior-agnostic estimation of discounted stationary distribution corrections. *Advances in neural information processing systems*, 32, 2019.

[50] Ofir Nachum, Bo Dai, Ilya Kostrikov, Yinlam Chow, Lihong Li, and Dale Schuurmans. Algaedice: Policy gradient from arbitrary experience. *arXiv preprint arXiv:1912.02074*, 2019.

[51] Gergely Neu, Anders Jonsson, and Vicenç Gómez. A unified view of entropy-regularized markov decision processes. *arXiv preprint arXiv:1705.07798*, 2017.

[52] Xinkun Nie and Stefan Wager. Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108(2):299–319, 2021.

[53] Antonio Olivas-Martinez and Andrea Rotnitzky. Source-condition analysis of kernel adversarial estimators. *arXiv preprint arXiv:2508.17181*, 2025.

[54] Andrew Patterson, Adam White, and Martha White. A generalized projected bellman error for off-policy value estimation in reinforcement learning. *Journal of Machine Learning Research*, 23(145):1–61, 2022.

[55] J Peng and R Williams. On the convergence of stochastic iterative dynamic programming algorithms. *Adaptive Behavior*, 1:437–454, 1993.

[56] Doina Precup, Richard S Sutton, and Satinder Singh. Eligibility traces for off-policy policy evaluation. In *Proceedings of the Seventeenth International Conference on Machine Learning*, pages 759–766. Morgan Kaufmann, 2000.

[57] Doina Precup, Richard S Sutton, and Sanjoy Dasgupta. Off-policy temporal-difference learning with function approximation. In *ICML*, pages 417–424, 2001.

[58] Martin L Puterman. Markov decision processes. *Handbooks in operations research and management science*, 2:331–434, 1990.

[59] Martin Riedmiller. Neural fitted q iteration–first experiences with a data efficient neural reinforcement learning method. In *European conference on machine learning*, pages 317–328. Springer, 2005.

[60] Kenneth Rose. Deterministic annealing for clustering, compression, classification, regression, and related optimization problems. *Proceedings of the IEEE*, 86(11):2210–2239, 2002.

[61] Bruno Scherrer. Approximate policy iteration schemes: A comparison. In *International Conference on Machine Learning*, pages 1314–1322. PMLR, 2014.

[62] Richard S Sutton, A Rupam Mahmood, and Martha White. An emphatic approach to the problem of off-policy temporal-difference learning. *Journal of Machine Learning Research*, 17 (73):1–29, 2016.

[63] Philip Thomas and Emma Brunskill. Data-efficient off-policy policy evaluation for reinforcement learning. In *International conference on machine learning*, pages 2139–2148. PMLR, 2016.

[64] Samuele Tosatto, Matteo Pirotta, Carlo d'Eramo, and Marcello Restelli. Boosted fitted q-iteration. In *International Conference on Machine Learning*, pages 3434–3443. PMLR, 2017.

[65] Hans Triebel. *Theory of function spaces III*. Springer, 2006.

[66] John Tsitsiklis and Benjamin Van Roy. Analysis of temporal-difference learning with function approximation. *Advances in neural information processing systems*, 9, 1996.

[67] Masatoshi Uehara, Jiawei Huang, and Nan Jiang. Minimax weight and Q-function learning for off-policy evaluation. In *International Conference on Machine Learning*, pages 9659–9668. PMLR, 2020.

[68] Masatoshi Uehara, Masaaki Imaizumi, Nan Jiang, Nathan Kallus, Wen Sun, and Tengyang Xie. Finite sample analysis of minimax offline reinforcement learning: Completeness, fast rates and first-order efficiency. *arXiv preprint arXiv:2102.02981*, 2021.

[69] Masatoshi Uehara, Nathan Kallus, Jason D Lee, and Wen Sun. Offline minimax Soft-Q-learning under realizability and partial coverage. *Advances in Neural Information Processing Systems*, 36:12797–12809, 2023.

[70] Sara van de Geer. On the uniform convergence of empirical norms and inner products, with application to causal inference. *Electronic Journal of Statistics*, 8:543–574, 2014.

[71] Lars van der Laan. A researcher's guide to empirical risk minimization. *arXiv preprint arXiv:2602.21501*, 2026.

[72] Lars van der Laan and Nathan Kallus. Fitted q evaluation without bellman completeness via stationary weighting. Manuscript, submitted to arXiv, 2025.

[73] Lars van der Laan, Nathan Kallus, and Aurélien Bibaut. Nonparametric instrumental variable inference with many weak instruments. *arXiv preprint arXiv:2505.07729*, 2025.

[74] Aad Van Der Vaart and Jon A Wellner. A local maximal inequality under uniform entropy. *Electronic Journal of Statistics*, 5(2011):192, 2011.

[75] Aad W Van Der Vaart and Jon A Wellner. Weak convergence. In *Weak convergence and empirical processes: with applications to statistics*, pages 16–28. Springer, 1996.

[76] Benjamin Van Roy. Performance loss bounds for approximate value iteration with state aggregation. *Mathematics of Operations Research*, 31(2):234–244, 2006.

[77] Cameron Voloshin, Hoang M Le, Nan Jiang, and Yisong Yue. Empirical study of off-policy policy evaluation for reinforcement learning. *arXiv preprint arXiv:1911.06854*, 2019.

[78] Martin J Wainwright. *High-dimensional statistics: A non-asymptotic viewpoint*, volume 48. Cambridge university press, 2019.

[79] Jiayi Wang, Zhengling Qi, and Raymond KW Wong. Projected state-action balancing weights for offline reinforcement learning. *The Annals of Statistics*, 51(4):1639–1665, 2023.

[80] Yuanhao Wang, Ruosong Wang, and Sham Kakade. An exponential lower bound for linearly realizable mdp with constant suboptimality gap. *Advances in Neural Information Processing Systems*, 34:9521–9533, 2021.

[81] Tengyang Xie and Nan Jiang. Batch value-function approximation with only realizability. In *International Conference on Machine Learning*, pages 11404–11413. PMLR, 2021.

[82] Tengyang Xie, Ching-An Cheng, Nan Jiang, Paul Mineiro, and Alekh Agarwal. Bellman-consistent pessimism for offline reinforcement learning. *Advances in neural information processing systems*, 34:6683–6694, 2021.

[83] Tengyang Xie, Dylan J Foster, Yu Bai, Nan Jiang, and Sham M Kakade. The role of coverage in online reinforcement learning. *arXiv preprint arXiv:2210.04157*, 2022.

[84] Lin Yang and Mengdi Wang. Reinforcement learning in feature space: Matrix bandit, kernels, and regret bound. In *International Conference on Machine Learning*, pages 10746–10756. PMLR, 2020.

[85] Huizhen Yu and Dimitri P Bertsekas. Weighted bellman equations and their applications in approximate dynamic programming. *Lab. for Information and Decision Systems Report LIDS-P-2876, MIT*, 2012.

[86] Huizhen Yu, A Rupam Mahmood, and Richard S Sutton. On generalized bellman equations and temporal-difference learning. *Journal of Machine Learning Research*, 19(48):1–49, 2018.

[87] Wenhao Zhan, Shicong Cen, Baihe Huang, Yuxin Chen, Jason D Lee, and Yuejie Chi. Policy mirror descent for regularized reinforcement learning: A generalized framework with linear convergence. *SIAM Journal on Optimization*, 33(2):1061–1091, 2023.

[88] Ruiyi Zhang, Bo Dai, Lihong Li, and Dale Schuurmans. Gendice: Generalized offline estimation of stationary values. *arXiv preprint arXiv:2002.09072*, 2020.

[89] Shangtong Zhang and Shimon Whiteson. Truncated emphatic temporal difference methods for prediction and control. *Journal of Machine Learning Research*, 23(153):1–59, 2022.

[90] Brian D. Ziebart. *Modeling Purposeful Adaptive Behavior With the Principle of Maximum Causal Entropy*. PhD thesis, Carnegie Mellon University, 2010.

[91] Brian D. Ziebart, Andrew Maas, J. Andrew Bagnell, and Anind K. Dey. Maximum entropy inverse reinforcement learning. In *AAAI*, 2008.

A Discussion and Conclusion

This paper identifies a simple mechanism behind the stability of fitted soft value iteration without Bellman completeness. Near the soft-optimal fixed point, the soft Bellman operator behaves like policy evaluation under the soft-optimal policy, whose natural contraction norm is the corresponding stationary state-action norm. Standard off-policy soft FQI instead projects Bellman targets in the behavior norm. Under function approximation and distribution shift, this projection-norm mismatch can destroy the local contraction that would otherwise stabilize fitted iteration.

Stationary reweighting addresses this mismatch without changing the Bellman target, adding an adversarial critic, or imposing Bellman completeness. It changes only the regression weights, so that the fitted projection approximates the stationary projection suggested by the local dynamics. Our finite-sample analysis shows that, inside a contraction basin, the iterates converge linearly to the stationary projected fixed point up to statistical and weight-estimation errors. The bound also clarifies how approximate weights enter: they affect the regression curvature and interact with the Bellman approximation residual, so imperfect weights are less harmful when the function class is closer to Bellman complete.

The result is local, and its scope is determined by basin entry, coverage, and weight estimation. Stationary weighting requires sufficient overlap between the behavior distribution and the relevant stationary distribution, and poor overlap can make exact ratios too variable. This motivates regularized, discounted, clipped, or otherwise stabilized occupancy weights in practice, as well as warm starts and temperature annealing as basin-entry strategies. These requirements reflect the usual coverage and stability challenges in off-policy value learning, rather than a limitation specific to stationary reweighting.

Stationary weighting should be viewed as complementary to minimax and adversarial Bellman-residual methods. Minimax objectives provide a powerful way to control Bellman residuals through critic classes and can perform well with suitable tuning. Stationary weighting offers a different route: it preserves the supervised-regression structure of FQI while aligning the fitted objective with the distribution under which the local Bellman dynamics contract. Understanding how these approaches can be combined, and how to tune them reliably under limited overlap, is an important direction for future work.

Overall, the main message is that Bellman completeness is not the only way to control fitted soft value iteration. When the projection norm is aligned with the stationary geometry of the target policy, local contraction can be recovered under misspecification. This perspective makes the role of coverage explicit, explains when ordinary soft FQI is already stable, and provides a modular path for stabilizing regression-based offline RL with existing density-ratio and balancing-weight tools.

B Main experiment results

B.1 Illustrative Garnet experiment

We generate Garnet MDPs [11] with $|\mathcal{S}| = 50$, $|\mathcal{A}| = 4$, branching factor 5, and $\gamma = 0.99$. The target is the near-hard-max soft-optimal policy at $\tau = 10^{-6}$, while the behavior policy is drawn independently at each state, creating severe norm mismatch. We use a five-dimensional linear class that contains $Q^\star_\tau$ but is not Bellman complete: in each seed, the first basis vector is the normalized $Q^\star_\tau$ table and the remaining four basis vectors are independent Gaussian state-action features orthogonalized against it. We collect 10^5 reset-sampled one-step transitions. Figure 2 shows the mechanism cleanly: under low temperature, behavior-distribution-norm FQI can be unstable; stationary weighting stabilizes the projected dynamics; and annealing improves the trajectory into the low-temperature regime.

B.2 Continuous offline RL

Expanded setup. We evaluate a continuous-state navigation MDP designed to separate Bellman incompleteness from off-policy distribution shift. The environment has five actions, nonlinear stochastic dynamics, a goal region, and a decoy region that attracts behavior data. A grid solve at $\tau = 10^{-3}$ gives $Q^\star_\tau$, the target stationary distribution, and oracle density ratios. Across ten behavior

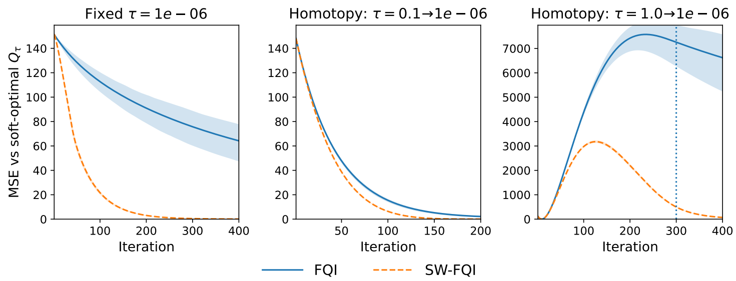

Figure 2: Illustrative Garnet diagnostic. Left: direct training at $\tau = 10^{-6}$. Middle/right: temperature homotopy. Shaded regions show 25%–75% quantiles over 200 random MDPs/datasets; the dashed line marks when τ reaches its target.

regimes, from nearly on-policy to severe shift, we run 60 iterations with a deliberately Bellman-incomplete linear state-action Q-class, comparing unweighted soft FQI, stabilized oracle stationary weighting, and estimated local RBF-polynomial stationary weighting. Both weighted methods clip and renormalize positive weights and mix toward uniform weights when effective sample size is too small. The estimated method uses a DICE-style saddle-point moment estimator with a shared RBF-polynomial dictionary and a discounted target ($\gamma_w = 0.95$) instead of the exact stationary target. We report direct low-temperature fitting at $\tau = 10^{-3}$; Appendix C gives details and annealing comparisons, which gave modest but consistent improvements.

Expanded discussion. Figure 1 In Appendix B.2 reports stationary projected Bellman error $\|Q - \mathcal{T}_{\mathcal{F}}(Q)\|_{2,\mu^\star}$, which vanishes at the projected fixed point $Q^\dagger$, and advantage-centered stationary Q-RMSE, $\|(Q - Q^\star_\tau) - \pi^\star(Q - Q^\star_\tau)\|_{2,\mu^\star}$, which removes constant shifts and captures policy-relevant value error. On policy, weighting is essentially neutral. As shift increases, unweighted FQI has larger target-stationary Bellman and advantage errors, while weighting reduces these errors in this controlled setting. Under severe shift, poor coverage makes exact stationary ratios highly variable, creating a bias–variance tradeoff between exact and stabilized approximate reweighting. Estimated local RBF-polynomial weights improve the policy-relevant quantities in most seeds: they beat unweighted FQI on advantage error in 407 of 500 seeds and policy value in 488 of 500 seeds. This does not uniformly translate to projected Bellman residual, where estimated weights beat unweighted FQI in 234 of 500 seeds. Thus stable approximate weighting can improve policy quality under limited coverage even when it does not uniformly reduce projected Bellman error.

The main figure for the continuous offline RL experiment is Figure 1.

B.3 Minimax Bellman-residual comparison

Expanded comparison with minimax Bellman-residual fitting. We compare stationary weighting with a regularized minimax Bellman-residual baseline using the same linear Q-features as soft FQI and the same RBF-polynomial critic dictionary used for ratio learning. This gives a strong residual-minimization comparator without changing the actor's approximation class. Figure 3 reports policy-relevant advantage error, unprojected target-stationary Bellman residual, and target-stationary Q-error, evaluating minimax on its natural residual metric and on the Q- and advantage-level quantities used in the main analysis. Estimated stationary weights track the stabilized oracle correction when coverage is adequate; minimax fitting is more competitive in severe-shift regimes where ratio estimation is difficult. Figure 5 and Table 1 in Appendix C show that heavier Tikhonov regularization can improve the minimax residual while leaving substantial target-stationary Q-error.

C Controlled simulation details

Environment and grid reference. The benchmark is a continuous two-dimensional navigation MDP with five discrete actions, nonlinear stochastic dynamics, a high-reward goal basin, and a lower-reward decoy basin. The decoy is used only to construct behavior policies whose stationary distributions differ from the soft-optimal stationary distribution. We compute reference quantities on a grid: soft value iteration with temperature $\tau = 10^{-3}$ gives $Q^\star_\tau$, the soft-optimal policy, and the

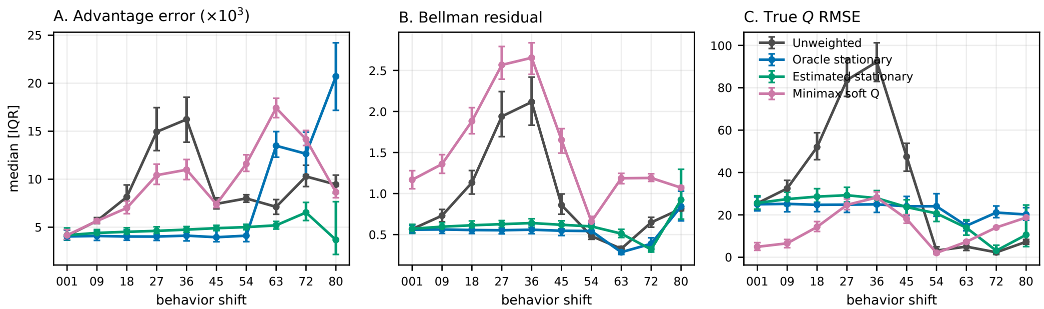

Figure 3: Minimax Bellman-residual comparison. We compare unweighted soft FQI, stabilized-oracle stationary weighting, estimated stationary weighting, and a Tikhonov-regularized minimax Bellman-residual baseline over the same ten behavior regimes. Points and intervals show medians and IQRs over 500 datasets. The panels report target-stationary advantage error, the unprojected target-stationary Bellman residual, and target-stationary Q-error.

corresponding stationary state-action distribution. Oracle density ratios are the soft-optimal stationary probabilities divided by the behavior stationary probabilities. The state space is the box $[-1,1]^2$, discretized as a 25×25 grid for reference dynamic programming. The five action vectors are $(0,0)$, $(0.23,0)$, $(-0.23,0)$, $(0,0.23)$, and $(0,-0.23)$. For state $s=(x,y)$ and action $a=(a_x,a_y)$, the transition mean before clipping to the box is

$$s + 0.13 \begin{pmatrix} \sin(\pi y) + 0.35x(1-y^2) \\ -0.85\sin(\pi x) + 0.25y(1-x^2) \end{pmatrix} + a + 0.055 \begin{pmatrix} a_y x \\ -a_x y \end{pmatrix}.$$

The grid transition probabilities are proportional to a Gaussian density around this mean with standard deviation 0.075, mixed with uniform teleportation probability 0.002. Rewards are evaluated at the transition mean:

$$\begin{aligned} r(s,a) &= 1.20\, b_{(0.62,0.62),0.20} + 0.55\, b_{(-0.55,0.45),0.23} - 0.22\, b_{(0.05,-0.10),0.22} \\ &\quad - 0.035\frac{\|a\|_2^2}{0.23^2} + 0.055\{\sin(2\pi x)a_x - \cos(\pi y)a_y\}, \end{aligned}$$

where $b_{c,\sigma}(z) = \exp\{-\|z-c\|_2^2/(2\sigma^2)\}$. Each finite-sample run uses $n = 5000$ one-step transitions, discount $\gamma = 0.97$, 60 FQI iterations, and ridge 10^{-4}. The main direct/annealed comparison uses 500 seeds, and the oracle annealing sensitivity uses 200 seeds.

Behavior regimes. Only the data-collection distribution changes across regimes. We use ten mixtures ranging from $0.999\pi^\star + 0.001\pi_{\text{unif}}$ to $0.20\pi^\star + 0.70\pi_{\text{decoy}} + 0.10\pi_{\text{unif}}$, with intermediate target-policy masses $0.91, 0.82, \ldots, 0.28$. The mild-shift regime, $0.55\pi^\star + 0.35\pi_{\text{decoy}} + 0.10\pi_{\text{unif}}$, has total variation distance 0.507, ESS fraction 0.428, and 99th percentile oracle ratio 1.5. The most severe regime has total variation distance 0.964, ESS fraction 0.017, and 99th percentile oracle ratio 34.

For each behavior regime, we report two overlap diagnostics. The total variation distance is

$$\mathrm{TV}(\mu^\star, \nu_b) = \frac{1}{2}\sum_{s,a} |\mu^\star(s,a) - \nu_b(s,a)|\,,$$

where $\mu^\star$ is the soft-optimal stationary state-action distribution and ν_b is the behavior stationary state-action distribution. The effective sample size fraction is the population oracle-ratio quantity

$$\mathrm{ESS} = \frac{1}{E_{\nu_b}[(d^\star)^2]}, \qquad d^\star = \frac{d\mu^\star}{d\nu_b},$$

the population analogue of $(\sum_i w_i)^2/(n\sum_i w_i^2)$.

Weight stabilization and estimated stationary weights. The stabilized oracle method starts from the grid-computed stationary density ratio to the fixed soft-optimal reference policy $\pi^\star_\tau$. In the experiments the weighted regressions use this fixed target/reference weight rather than refitting a new ratio for every FQI iterate. The estimated local RBF-polynomial method uses the same state-action samples as FQI. The linear Q-class has ten features

$$1,\ x,\ y,\ a_x,\ a_y,\ x^2,\ y^2,\ a_x^2+a_y^2,\ b_{(0.62,0.62),0.28}(s),\ b_{(-0.55,0.45),0.30}(s).$$

The ratio and critic share a finite RBF-polynomial dictionary consisting of ten polynomial state-action features plus 36 state centers crossed with five actions, for 190 total features. The RBF bandwidth is 0.65 times the median positive distance between dictionary centers. The DICE-style estimator uses primal and dual ridge penalties 10^{-5} and normalization penalty 10. We target discounted stationary ratios with $\gamma_w = 0.95$, which regularizes the stationary-flow equations while keeping the weights close to the stationary correction. For both weighted methods, raw weights are truncated below at 10^{-8}, clipped at the smaller of the empirical 99th percentile and 25, renormalized to have mean one, and mixed as $(1-\lambda)w + \lambda$. We choose $\lambda \in [0, 0.50]$ by binary search to raise the empirical ESS fraction to at least 0.25 when possible, using $\lambda = 0.50$ otherwise.

Diagnostics. Stationary projected Bellman error is the Bellman residual after projection in the target stationary norm. Advantage-centered stationary Q-error subtracts the within-state action average before computing target-stationary error, removing raw Q-shifts that do not affect low-temperature action choice. Panel D reports severe-shift win rates against unweighted FQI for the three diagnostics shown or discussed in the main text.

Minimax Bellman-residual baseline. The minimax comparison uses the same Q-function class as the corresponding soft FQI run. For a linear parameterization Q_θ, let $\delta_\theta(S, A, R, S')$ denote the soft Bellman residual formed with the same target update and temperature as the fitted-Q regressions. The critic class is the same RBF-polynomial dictionary $\psi(S, A)$ used by the stationary-ratio estimator. The baseline solves the regularized empirical projected-residual objective

$$\min_\theta \|\mathbb{P}_n \psi\, \delta_\theta\|^2_{(\mathbb{P}_n \psi\psi^\top + \lambda_c I)^{-1}} + \lambda_q \|\theta\|_2^2,$$

with $\lambda_q = \lambda_c = 10^{-4}$ unless otherwise stated. We also ran a ridge-sensitivity check over

$$10^{-6}, 3\cdot 10^{-6}, 10^{-5}, 3\cdot 10^{-5}, 10^{-4}, 3\cdot 10^{-4}, 10^{-3}, 3\cdot 10^{-3}, 10^{-2}, 3\cdot 10^{-2}, 10^{-1},$$

and a cross-validated ridge choice using the same ridge family as the ratio estimator. The estimated-weight ablation likewise includes fixed-ridge and cross-validated Tikhonov choices. Thus the comparison does not give stationary weighting a larger Q-class; it compares two ways of stabilizing fitted Bellman updates, either by changing the regression geometry through stationary weights or by controlling Bellman moments with an adversarial residual critic.

Additional stress tests. We include three diagnostics to separate the proposed mechanism from generic regularization effects. First, a rich-feature negative control augments the linear action features with RBF state-action features, substantially reducing projection mismatch. In the mild and moderate regimes, the gains from stationary weighting become modest, as expected if weighting mainly corrects the projection geometry rather than acting as a generic regularizer; the severe-shift case remains coverage-limited. Second, a sample-size sweep varies the number of behavior transitions from 1000 to 10000 in three shift regimes. The estimated-weight gap to oracle weighting shrinks as effective sample size improves, while the most severe regimes remain limited by coverage. Third, a weight-estimator ablation varies the discounted stationary target $\gamma_w \in \{0, 0.5, 0.95, 1\}$ and compares fixed-ridge and cross-validated ridge choices. The ablation shows the expected bias–variance tradeoff: nearly stationary targets can be accurate when coverage is adequate, while discounted or ridge-stabilized targets are more reliable under severe shift.

Oracle tuning check. Table 1 gives a deliberately generous tuning check. For each family and regime, we select the Tikhonov setting, and for estimated weights also the stationary target γ_w, that minimizes the target-stationary Bellman residual after observing the true grid reference. This is not a deployable model-selection rule; it is a sensitivity analysis to check that the comparison is not an artifact of the default ridge. Oracle tuning improves residuals for both families in some regimes, but heavily regularized minimax fits can still have large true-Q error. The qualitative pattern is unchanged: minimax remains a useful Bellman-residual baseline, while stationary weighting more directly targets the target-stationary Q geometry.

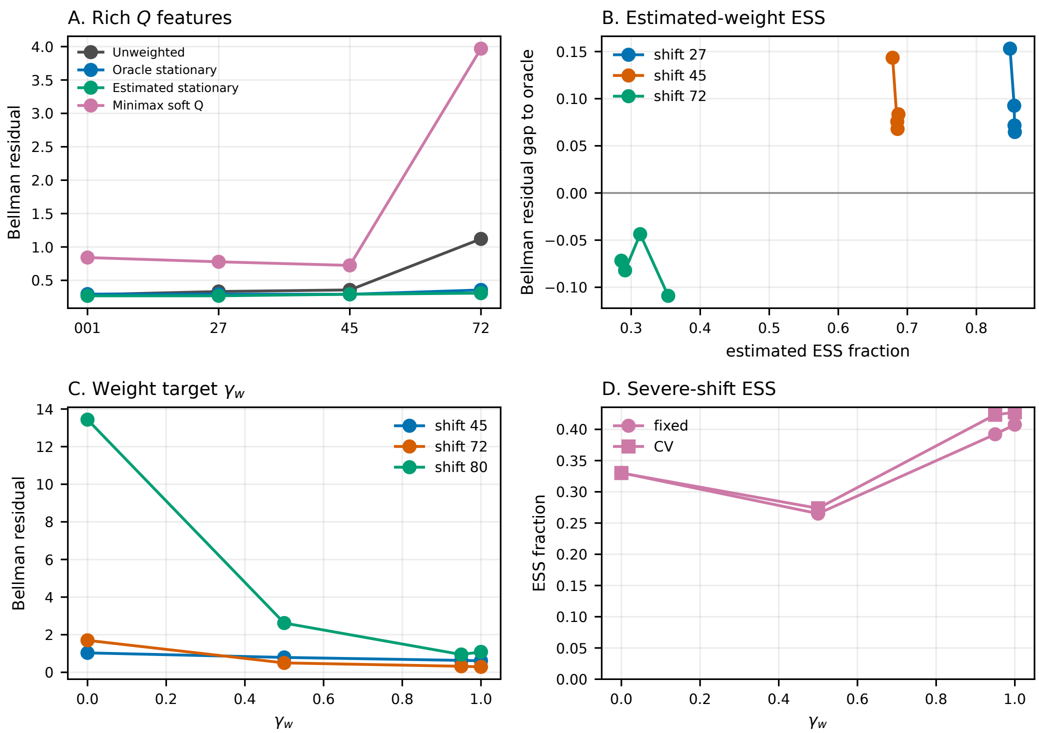

Figure 4: **Supporting diagnostics.** Richer Q-features reduce the weighting gap in the mild and moderate regimes, while the severe-shift case remains coverage-limited. The ESS and γ_w sweeps show when estimated stationary weights track the oracle correction and when finite-sample weight error dominates.

Table 1: Oracle-tuned Tikhonov sensitivity for estimated stationary weights and minimax soft Q. Entries are medians with interquartile ranges over the corresponding sweep. The oracle criterion column indicates which metric was used to choose the tuning rule within each family and regime.

family	regime	oracle criterion	selected Tikhonov/target	Bellman RMSE	Advantage x1e3	True Q RMSE	Value error
estimated weights	shift 45	Bellman RMSE	$\gamma_w = 1$, fixed, $\lambda = 1e-05$	0.614 [0.555, 0.654]	4.97 [4.69, 5.24]	22.1 [18.8, 25.6]	25.6 [25.6, 25.6]
estimated weights	shift 72	Bellman RMSE	$\gamma_w = 1$, fixed, $\lambda = 1e-05$	0.291 [0.27, 0.326]	6.55 [5.67, 7.51]	2.49 [1.22, 4.86]	25.6 [25.6, 27.3]
estimated weights	shift 80	Bellman RMSE	$\gamma_w = 0.95$, fixed, $\lambda = 1e-05$	0.946 [0.715, 1.21]	3.32 [2.36, 6.91]	10.6 [4.45, 20.7]	25.6 [25.6, 27.1]
minimax soft Q	shift 27	Bellman RMSE	$\lambda_q = \lambda_c = 10^{-1}$	0.953 [0.951, 0.956]	0.282 [0.279, 0.287]	20.3 [20.1, 20.4]	25.6 [25.6, 25.6]
minimax soft Q	shift 45	Bellman RMSE	$\lambda_q = \lambda_c = 10^{-1}$	0.903 [0.901, 0.905]	0.222 [0.215, 0.226]	22.2 [22.1, 22.3]	25.6 [25.6, 25.6]
minimax soft Q	shift 72	Bellman RMSE	$\lambda_q = \lambda_c = 10^{-2}$	0.786 [0.782, 0.792]	3.02 [2.94, 3.11]	21.1 [20.7, 21.4]	25.6 [25.6, 25.6]
minimax soft Q	shift 80	Bellman RMSE	$\lambda_q = \lambda_c = 3 \times 10^{-2}$	0.891 [0.889, 0.893]	1.02 [0.99, 1.04]	26.7 [26.7, 26.8]	27.3 [27.3, 27.3]

Annealing sensitivity. The main continuous experiment uses direct low-temperature fitting at $\tau = 10^{-3}$. To check whether continuation becomes more important near the hard-max limit, we also repeat the experiment at $\tau = 10^{-6}$. For $\tau = 10^{-3}$, we use the staged schedule $0.2, 0.05, 0.01, 0.003, 10^{-3}$; for $\tau = 10^{-6}$, we append a final 10^{-6} stage. Table 2 reports median stationary projected Bellman error over 200 datasets. Annealing has a small effect at both temperatures: the largest median change is 1.26% at $\tau = 10^{-3}$ and 1.21% at $\tau = 10^{-6}$.

D Local Geometry and Operator Smoothness

This appendix collects the local contraction picture and the operator-smoothness facts used to prove the local contraction theorem.

D.1 Local contraction geometry

Figure 7 summarizes the geometry behind the local contraction argument. Inside $\mathbb{B}_{\mathcal{F}}(r_{\text{loc}}, Q^\star)$, the Bellman map $\mathcal{T}$ contracts toward $Q^\star$, while the projection $\Pi_{\mathcal{F}}$ keeps iterates in $\mathcal{F}$. Starting in

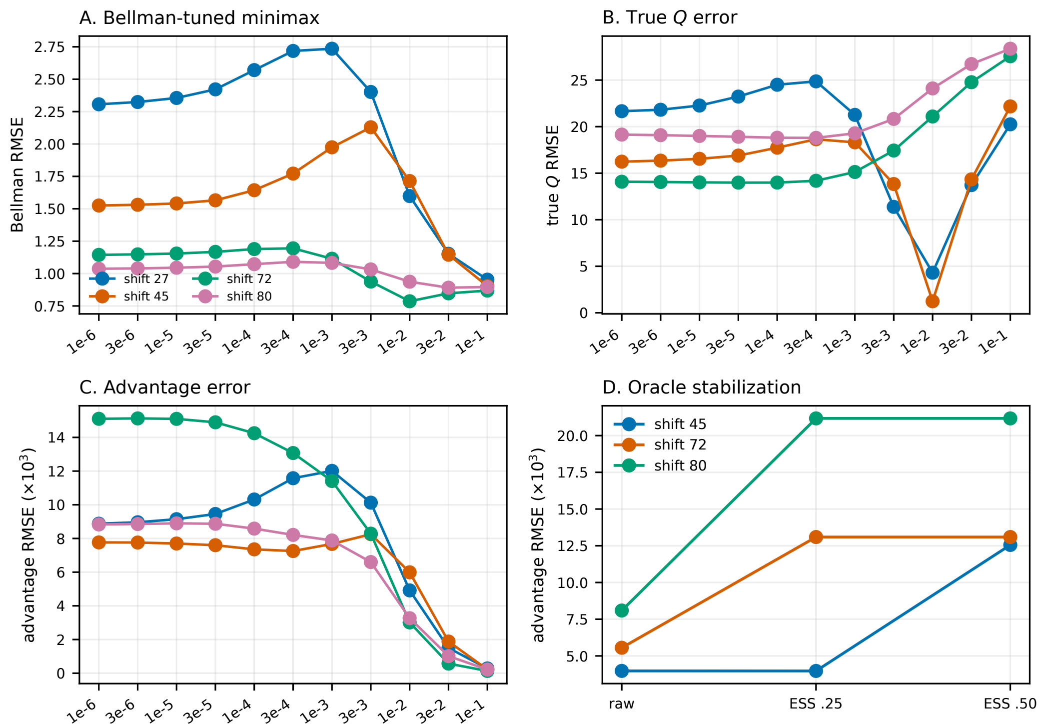

Figure 5: **Minimax and oracle-stabilization sensitivity.** Varying the minimax ridge reveals the residual–shrinkage tradeoff: stronger regularization can reduce Bellman residuals and advantage error while still leaving large target-stationary Q-error. Oracle stabilization controls the severe-shift bias–variance tradeoff for exact stationary ratios.

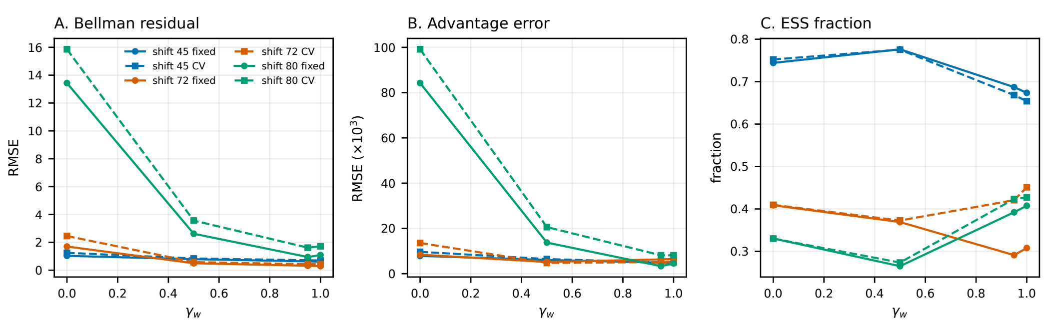

Figure 6: **Estimated-weight sensitivity.** We vary the stationary target γ_w and compare fixed versus cross-validated Tikhonov regularization for the density-ratio estimator. The severe-shift regimes show the expected bias–variance tradeoff: more target-stationary weights improve the residual when coverage is adequate, while ESS and ridge selection determine stability.

$\mathbb{B}_{\mathcal{F}}(r, Q^\dagger)$ with $r + \varepsilon_{\mathcal{F}} < r_{\mathrm{loc}}$, the iterates remain inside the contraction region and move toward the constrained fixed point $Q^\dagger$.

D.2 Operator smoothness results

D.2.1 First derivative of the Bellman operator

Lemma 4 (First derivative of the soft Bellman operator)**.** *For any Q and perturbation H, the Fréchet derivative of $\mathcal{T}$ at Q is*

$$D\mathcal{T}(Q)[H] \;=\; \gamma\, P^{\mathrm{eval}}_{\pi_Q} H.$$

Table 2: Annealing sensitivity. Entries are median stationary projected Bellman error over 200 offline datasets. Gain is $(\text{direct} - \text{annealed})/\text{direct}$.

τ	Regime	Method	Direct	Annealed	Gain
10^{-3}	On-policy	Unweighted	0.526	0.521	0.94%
10^{-3}	On-policy	Stab. oracle	0.516	0.510	1.05%
10^{-3}	On-policy	Estimated	0.533	0.527	1.05%
10^{-3}	Mild shift	Unweighted	0.822	0.819	0.37%
10^{-3}	Mild shift	Stab. oracle	0.510	0.504	1.10%
10^{-3}	Mild shift	Estimated	0.584	0.577	1.17%
10^{-3}	Severe shift	Unweighted	0.798	0.788	1.26%
10^{-3}	Severe shift	Stab. oracle	0.804	0.797	0.80%
10^{-3}	Severe shift	Estimated	0.839	0.832	0.82%
10^{-6}	On-policy	Unweighted	0.529	0.524	0.94%
10^{-6}	On-policy	Stab. oracle	0.519	0.515	0.80%
10^{-6}	On-policy	Estimated	0.533	0.529	0.79%
10^{-6}	Mild shift	Unweighted	0.811	0.808	0.38%
10^{-6}	Mild shift	Stab. oracle	0.508	0.504	0.85%
10^{-6}	Mild shift	Estimated	0.581	0.574	1.09%
10^{-6}	Severe shift	Unweighted	0.798	0.789	1.21%
10^{-6}	Severe shift	Stab. oracle	0.815	0.809	0.83%
10^{-6}	Severe shift	Estimated	0.849	0.845	0.42%

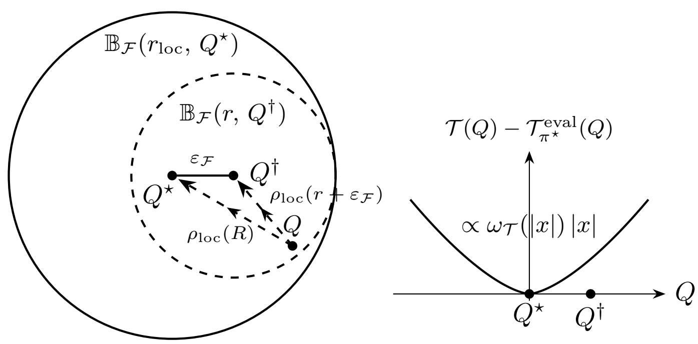

Figure 7: Contraction geometry. Left: contraction region $\mathbb{B}_{\mathcal{F}}(r_{\text{loc}}, Q^\star)$ and basin of attraction $\mathbb{B}_{\mathcal{F}}(r, Q^\dagger)$, where $r + \varepsilon_{\mathcal{F}} < r_{\text{loc}}$. Right: one–dimensional curvature slice of $\mathcal{T}(Q) - \mathcal{T}^{\text{eval}}_{\pi^\star}(Q)$ near $Q^\star$, with distances $R = \text{dist}(Q, Q^\star)$ and $r = \text{dist}(Q, Q^\dagger)$.

In particular, at $Q^\star$, $\|D\mathcal{T}(Q^\star)[H]\|_{2,\mu^\star} \ \leq\ \gamma\,\|H\|_{2,\mu^\star}$.

Proof of Lemma 4. Fix (s, a) and a perturbation H. Since the reward term does not depend on Q,

$$D\mathcal{T}(Q)[H](s,a) = \gamma\,\mathbb{E}_{S' \sim P(\cdot|s,a)}\big[D\phi_{S'}(Q)[H]\big],$$

where $\phi_{S'}(Q) := \tau \log \sum_b \exp(Q(S', b)/\tau)$. Let $g(z) = \tau \log \sum_b e^{z_b/\tau}$ with $z_b(Q) = Q(S', b)$. The gradient of g is the softmax, so $\partial g/\partial z_b(z(Q)) = \pi_Q(b \mid S')$. By the chain rule,

$$D\phi_{S'}(Q)[H] = \sum_b \pi_Q(b \mid S')\, H(S', b).$$

Substituting back gives

$$\begin{aligned} D\mathcal{T}(Q)[H](s,a) &= \gamma\,\mathbb{E}_{S' \sim P(\cdot|s,a)}\Big[\sum_b \pi_Q(b \mid S')\, H(S', b)\Big] \\ &= \gamma\, P^{\text{eval}}_{\pi_Q} H(s,a). \end{aligned}$$

This proves the claim. □

D.2.2 Second derivative and local curvature

Computation of second derivative. The next lemma shows that the second derivative of $\mathcal{T}$ has a positive–semidefinite covariance form.

Lemma 5 (Second derivative)**.** *For any Q and directions H_1, H_2,*

$$D^2\mathcal{T}(Q)[H_1, H_2](s,a) = \\ \gamma\, \mathbb{E}_{S'\sim P(\cdot|s,a)} \left[\tfrac{1}{\tau} \operatorname{Cov}_{\pi_Q}\big(H_1(S',\cdot),\ H_2(S',\cdot)\big)\right].$$

Proof of Lemma 5. Fix (s,a) and directions H_1, H_2. As in Lemma 4,

$$(\mathcal{T}Q)(s,a) = r_0(s,a) + \gamma\, \mathbb{E}_{S'\sim P(\cdot|s,a)}[\phi_{S'}(Q)],$$

$$\phi_{S'}(Q) := \tau \log \sum_b \exp\!\big(Q(S',b)/\tau\big).$$

Thus,

$$D^2\mathcal{T}(Q)[H_1, H_2](s,a) \\ = \gamma\, \mathbb{E}_{S'\sim P(\cdot|s,a)}\big[D^2\phi_{S'}(Q)[H_1, H_2]\big].$$

For fixed S', write $z_b(Q) = Q(S',b)$ and $g(z) = \tau \log \sum_b e^{z_b/\tau}$, so $\phi_{S'}(Q) = g(z(Q))$. Let $\pi_Q(\cdot \mid S')$ be the softmax at $z(Q)$. Then

$$\frac{\partial g}{\partial z_b}\big(z(Q)\big) = \pi_Q(b \mid S'),$$

$$\frac{\partial^2 g}{\partial z_b\, \partial z_c}\big(z(Q)\big) = \frac{1}{\tau}\Big[\pi_Q(b \mid S')\, \mathbf{1}\{b = c\} \\ -\ \pi_Q(b \mid S')\, \pi_Q(c \mid S')\Big].$$

Hence

$$\begin{aligned} &D^2\phi_{S'}(Q)[H_1, H_2] \\ &\quad= \sum_{b,c} \frac{\partial^2 g}{\partial z_b\, \partial z_c}(z(Q))\, H_1(S',b)\, H_2(S',c) \\ &\quad= \frac{1}{\tau}\Big(\mathbb{E}_{\pi_Q}[H_1(S',\cdot)\, H_2(S',\cdot)] \\ &\qquad\qquad -\ \mathbb{E}_{\pi_Q}[H_1(S',\cdot)]\ \mathbb{E}_{\pi_Q}[H_2(S',\cdot)]\Big) \\ &\quad= \frac{1}{\tau} \operatorname{Cov}_{\pi_Q}\!\big(H_1(S',\cdot),\ H_2(S',\cdot)\big). \end{aligned}$$

Taking expectation over S' yields the claimed expression.

For positive semidefiniteness, set $H_1 = H_2 = H$:

$$D^2\mathcal{T}(Q)[H,H](s,a) = \frac{\gamma}{\tau}\, \mathbb{E}_{S'\sim P(\cdot|s,a)}\Big[\operatorname{Var}_{\pi_Q}\big(H(S',\cdot)\big)\Big] \\ \geq\ 0.$$

In particular, $D^2\mathcal{T}(Q)$ is positive semidefinite for every Q. Moreover, for any fixed S' the covariance form is strictly positive whenever $H(S',\cdot)$ is nonconstant under $\pi_Q(\cdot \mid S')$, so $D^2\mathcal{T}(Q)$ is strictly positive definite along nontrivial action-differential directions. □

A sufficient interpolation condition. The main theorem uses the intrinsic curvature modulus $\omega_{\mathcal{T}}$. The following stronger interpolation condition is a simple way to verify it in standard classes.

S1) L^∞–L^2 **control.** There exist $\alpha \in (0,1]$ and $C_\infty < \infty$ such that

$$\|H\|_\infty \leq C_\infty \|H\|_{2,\mu^\star}^\alpha$$

for all $H \in \mathcal{H}_{\mathcal{F}}^\star$.

Uniform boundedness of second derivative and curvature bound.

Lemma 6 (Uniform boundedness of the second derivative)**.** *Assume Conditions C1–S1 hold. Then there exists a finite constant*

$$\beta_{\mathrm{loc}} := \frac{\gamma}{2\tau}\, C_\infty \sqrt{\frac{|\mathcal{A}|}{\pi_{\min}}}$$

such that, for all Q, all $H_2 \in L^2(\mu^\star)$, and all $H_1 = tH$ with $H \in \mathcal{H}_{\mathcal{F}}^\star$ and $t \in [0,1]$,

$$\left\| D^2\mathcal{T}(Q)[H_1, H_2] \right\|_{2,\mu^\star} \;\le\; \beta_{\mathrm{loc}}\, \|H_1\|_{2,\mu^\star}^{\alpha}\, \|H_2\|_{2,\mu^\star}, \tag{3}$$

where α is the exponent from S1.

Proof. By Lemma 5, $D^2\mathcal{T}(Q)[H_1, H_2]$ can be expressed in terms of the covariance $\mathrm{Cov}_{\pi_Q}\big(H_1(S',\cdot),\, H_2(S',\cdot)\big)$.

For fixed S', write $h_i := H_i(S',\cdot) \in \mathbb{R}^{|\mathcal{A}|}$ and $p_Q := \pi_Q(\cdot \mid S')$. Then

$$\mathrm{Cov}_{\pi_Q}\big(H_1(S',\cdot), H_2(S',\cdot)\big) = h_1^\top \big(\mathrm{diag}(p_Q) - p_Q p_Q^\top\big) h_2,$$

and the covariance matrix $\mathrm{diag}(p_Q) - p_Q p_Q^\top$ has spectral norm at most $1/2$, so

$$\left|\mathrm{Cov}_{\pi_Q}\big(H_1(S',\cdot), H_2(S',\cdot)\big)\right| \;\le\; \tfrac{1}{2}\, \|h_1\|_2\, \|h_2\|_2.$$

Thus, for each (s,a),

$$\begin{aligned} &\left| D^2\mathcal{T}(Q)[H_1, H_2](s,a) \right| \\ &\quad\le\; \frac{\gamma}{2\tau}\, \mathbb{E}_{S'|s,a}\big[\, \|H_1(S',\cdot)\|_2\, \|H_2(S',\cdot)\|_2 \,\big]. \end{aligned}$$

Let $(S,A) \sim \mu^\star$ and $S' \sim P(\cdot \mid S,A)$, and define $Z(S') := \|H_1(S',\cdot)\|_2\, \|H_2(S',\cdot)\|_2$. By Jensen's inequality,

$$\begin{aligned} \left\| D^2\mathcal{T}(Q)[H_1, H_2] \right\|_{2,\mu^\star}^2 &= \mathbb{E}_{\mu^\star}\Big[D^2\mathcal{T}(Q)[H_1, H_2](S,A)^2 \Big] \\ &\le \Big(\frac{\gamma}{2\tau}\Big)^2\, \mathbb{E}_{\mu^\star}\Big[\mathbb{E}_{S'|S,A} Z(S')^2 \Big]. \end{aligned}$$

Since $\mu^\star$ is stationary for $\pi^\star$, the marginal distribution of S' under $(S,A) \sim \mu^\star$, $S' \sim P(\cdot \mid S,A)$ coincides with the state marginal of $\mu^\star$. Hence

$$\mathbb{E}_{\mu^\star}\Big[\mathbb{E}_{S'|S,A} Z(S')^2 \Big] = \mathbb{E}_{S'}\big[\|H_1(S',\cdot)\|_2^2\, \|H_2(S',\cdot)\|_2^2 \big].$$

By the assumption $\pi^\star(a \mid s) \ge \pi_{\min}$ whenever $\mu^\star(s,a) > 0$, for any state s and $h \in \mathbb{R}^{|\mathcal{A}|}$,

$$\begin{aligned} \|h\|_2^2 &= \sum_a h(a)^2 \\ &= \sum_a \frac{h(a)^2}{\pi^\star(a \mid s)}\, \pi^\star(a \mid s) \\ &\le \frac{1}{\pi_{\min}} \sum_a \pi^\star(a \mid s)\, h(a)^2. \end{aligned}$$

Applying this with $h = H_i(S',\cdot)$ gives

$$\|H_i(S',\cdot)\|_2^2 \;\le\; \frac{1}{\pi_{\min}} \sum_a \pi^\star(a \mid S')\, H_i(S',a)^2.$$

Since $H_1 = tH$ with $H \in \mathcal{H}^\star_{\mathcal{F}}$ and $t \in [0,1]$, Condition S1 and $t \le t^\alpha$ give $\|H_1\|_\infty \le C_\infty \|H_1\|^\alpha_{2,\mu^\star}$. Thus, using $\|H_1(S',\cdot)\|_2^2 \le \sup_s \|H_1(s,\cdot)\|_2^2 \le |\mathcal{A}|\, \|H_1\|_\infty^2$, we obtain

$$\begin{aligned}
&\mathbb{E}_{S'}\left[\|H_1(S',\cdot)\|_2^2\, \|H_2(S',\cdot)\|_2^2\right] \\
&\quad\le \sup_s \|H_1(s,\cdot)\|_2^2\, \mathbb{E}_{S'}\left[\|H_2(S',\cdot)\|_2^2\right] \\
&\quad\le |\mathcal{A}|\, \|H_1\|_\infty^2\, \mathbb{E}_{S'}\left[\|H_2(S',\cdot)\|_2^2\right] \\
&\quad\le \frac{|\mathcal{A}|}{\pi_{\min}}\, \|H_1\|_\infty^2\, \|H_2\|_{2,\mu^\star}^2 \\
&\quad\le \frac{|\mathcal{A}|}{\pi_{\min}}\, C_\infty^2\, \|H_1\|_{2,\mu^\star}^{2\alpha}\, \|H_2\|_{2,\mu^\star}^2.
\end{aligned}$$

Combining with the previous display gives

$$\begin{aligned}
&\left\|D^2\mathcal{T}(Q)[H_1,H_2]\right\|_{2,\mu^\star}^2 \\
&\quad\le\ \frac{|\mathcal{A}|}{\pi_{\min}}\, C_\infty^2 \left(\frac{\gamma}{2\tau}\right)^2 \|H_1\|_{2,\mu^\star}^{2\alpha}\, \|H_2\|_{2,\mu^\star}^2 \\
&\quad=\ \frac{|\mathcal{A}|\, C_\infty^2}{\pi_{\min}} \left(\frac{\gamma}{2\tau}\right)^2 \|H_1\|_{2,\mu^\star}^{2\alpha}\, \|H_2\|_{2,\mu^\star}^2.
\end{aligned}$$

so that

$$\left\|D^2\mathcal{T}(Q)[H_1,H_2]\right\|_{2,\mu^\star}\ \le\ \beta_{\mathrm{loc}}\, \|H_1\|_{2,\mu^\star}^{\alpha}\, \|H_2\|_{2,\mu^\star},$$

where $\beta_{\mathrm{loc}} := \frac{\gamma}{2\tau}\, C_\infty \sqrt{\frac{|\mathcal{A}|}{\pi_{\min}}}$. This proves the claimed bound.

For one-point Taylor remainders, Condition S1 can often be weakened to an L^4–L^2 coupling such as $\|H\|_{4,\mu^\star}^2 \le C_4 \|H\|_{2,\mu^\star}^{1+\alpha}$. For the pairwise contraction theorem, however, the needed sufficient condition is the bilinear derivative-drift bound in (3), since the second derivative must remain linear in the second direction. $\square$

Lemma 7 (Interpolation implies the curvature modulus)**.** *Assume Conditions C1 and S1. Then Condition C3 holds with*

$$\omega_{\mathcal{T}}(R) \le \beta_{\mathrm{loc}} R^\alpha, \qquad \beta_{\mathrm{loc}} := \frac{\gamma}{2\tau} C_\infty \sqrt{\frac{|\mathcal{A}|}{\pi_{\min}}}.$$

Consequently the polynomial sufficient radius

$$r_{\mathrm{poly}} := \left(\frac{1-\gamma}{\beta_{\mathrm{loc}}}\right)^{1/\alpha}$$

satisfies $r_{\mathrm{poly}} \le r_{\mathrm{loc}}$.

Proof. Let $Q \in \mathbb{S}^\star_{\mathcal{F}}(R)$ and $H \in \mathcal{H}^\star_{\mathcal{F}}$. By the mean-value representation,

$$\{D\mathcal{T}(Q) - D\mathcal{T}(Q^\star)\}[H] = \int_0^1 D^2\mathcal{T}(Q^\star + u(Q-Q^\star))[Q-Q^\star, H]\, du.$$

Since $Q \in \mathbb{S}^\star_{\mathcal{F}}(R)$, the direction $Q - Q^\star$ has the form $t(f - Q^\star)$ for some $f \in \mathcal{F}$ and $t \in [0,1]$. Applying Lemma 6 gives

$$\|(D\mathcal{T}(Q) - D\mathcal{T}(Q^\star))[H]\|_{2,\mu^\star} \le \beta_{\mathrm{loc}} \|Q - Q^\star\|_{2,\mu^\star}^\alpha \|H\|_{2,\mu^\star} \le \beta_{\mathrm{loc}} R^\alpha \|H\|_{2,\mu^\star}.$$

Taking the supremum over Q and H gives the claim. $\square$

D.3 Curvature calibrations and sup-norm local radii

Lemma 8 (Curvature calibrations)**.** *The following sufficient bounds hold.*

(i) If Condition S1 holds, then

$$\omega_{\mathcal{T}}(R) \le \beta_{\mathrm{loc}} R^{\alpha}, \qquad \beta_{\mathrm{loc}} := \frac{\gamma C_{\infty}}{2\tau}\sqrt{\frac{|\mathcal{A}|}{\pi_{\min}}}.$$

Thus

$$r_{\mathrm{loc}} \ge \left(\frac{1-\gamma}{\beta_{\mathrm{loc}}}\right)^{1/\alpha}.$$

(ii) Under the action-gap conditions Assumptions S1–S4,

$$\bar{\omega}_{\mathcal{T},\tau}(R) \le \bar{\beta}_{\mathrm{loc}}^{\mathrm{gap}}(\tau) R^{\alpha}, \qquad \bar{\beta}_{\mathrm{loc}}^{\mathrm{gap}}(\tau) := \frac{\gamma}{\tau} C_{\mathrm{ref}} C_{\mathrm{gap}} e^{-\Delta/\tau}.$$

Consequently, the fixed-reference gap analogue has radius at least

$$\bar{r}_{\mathrm{loc}}^{\mathrm{gap}}(\tau) := \min\left\{ r_{\mathrm{gap}},\ \left(\frac{1-\gamma\kappa_{\tau}}{\bar{\beta}_{\mathrm{loc}}^{\mathrm{gap}}(\tau)}\right)^{1/\alpha} \right\}.$$

Proof of Lemma 8. The first claim is Lemma 7. For the gap claim, define the fixed-reference analogue $\bar{\omega}_{\mathcal{T},\tau}$ by replacing $\|\cdot\|_{2,\mu^{\star}}$ with $\|\cdot\|_{2,\bar{\mu}}$ in Condition C3 and restricting the segment region to $\mathbb{B}_{\mathrm{gap}}$. Lemma 15 and the same mean-value argument as in Lemma 7 give

$$\bar{\omega}_{\mathcal{T},\tau}(R) \le \bar{\beta}_{\mathrm{loc}}^{\mathrm{gap}}(\tau) R^{\alpha}.$$

Solving $\gamma + \beta_{\mathrm{loc}} R^{\alpha} < 1$ gives the first radius bound. Solving $\gamma\kappa_{\tau} + \bar{\beta}_{\mathrm{loc}}^{\mathrm{gap}}(\tau) R^{\alpha} < 1$, while also requiring the segment to remain in $\mathbb{B}_{\mathrm{gap}}$, gives the fixed-reference gap radius in (17). $\square$

D.3.1 Sup-norm sufficient local radii

Under the sup-norm interpolation route, the abstract approximate-realizability condition has the following explicit sufficient form. Let

$$d_{\mathcal{F}} := \mathrm{dist}(\mathcal{F}, Q^{\star}), \qquad r_{\mathrm{poly}} := \left(\frac{1-\gamma}{\beta_{\mathrm{loc}}}\right)^{1/\alpha}.$$

If there exists $R_{\mathrm{fp}} \in (0, r_{\mathrm{poly}})$ such that

$$d_{\mathcal{F}} \le \{1 - \gamma - \beta_{\mathrm{loc}} R_{\mathrm{fp}}^{\alpha}\} R_{\mathrm{fp}},$$

then Condition C4 holds with this R_{fp}. Equivalently, it is enough that

$$d_{\mathcal{F}} \le \frac{\alpha}{\alpha+1}(1-\gamma)\left\{\frac{1-\gamma}{(\alpha+1)\beta_{\mathrm{loc}}}\right\}^{1/\alpha},$$

obtained by maximizing the right-hand side over R_{fp}. In this case $\varepsilon_{\mathcal{F}} \le R_{\mathrm{fp}}$, so any

$$0 < r < r_{\mathrm{poly}} - R_{\mathrm{fp}}$$

is an admissible initialization radius for Theorem 2; more generally, any r with $r + \varepsilon_{\mathcal{F}} < r_{\mathrm{poly}}$ is admissible, and its contraction rate satisfies

$$\rho_{\mathrm{loc}}(r + \varepsilon_{\mathcal{F}}) \le \gamma + \beta_{\mathrm{loc}}(r + \varepsilon_{\mathcal{F}})^{\alpha}.$$

D.3.2 Proof of the Bellman linearization remainder

Proof of Lemma 1. Fix $Q \in \mathbb{B}_{\mathcal{F}}(R, Q^\star)$, set $H := Q - Q^\star$, and define $Q_t := Q^\star + tH$. Then $Q_t \in \mathbb{S}^\star_{\mathcal{F}}(R)$ for $t \in [0,1]$. The fundamental theorem of calculus gives

$$\mathcal{T}(Q) - \mathcal{T}(Q^\star) = \int_0^1 D\mathcal{T}(Q_t)[H]\, dt.$$

By Lemma 4,

$$D\mathcal{T}(Q^\star)[Q - Q^\star] = \gamma\, P^{\mathrm{eval}}_{\pi^\star}(Q - Q^\star).$$

Since $Q^\star$ is a fixed point, $\mathcal{T}(Q^\star) = Q^\star$, and

$$\mathcal{T}(Q^\star) + D\mathcal{T}(Q^\star)[Q - Q^\star] = Q^\star + \gamma P^{\mathrm{eval}}_{\pi^\star}(Q - Q^\star) = \mathcal{T}^{\mathrm{eval}}_{Q^\star}(Q),$$

we have

$$\mathcal{T}(Q) - \mathcal{T}^{\mathrm{eval}}_{Q^\star}(Q) = \int_0^1 \{D\mathcal{T}(Q_t) - D\mathcal{T}(Q^\star)\}[H]\, dt.$$

Since $H \in \mathcal{H}^\star_{\mathcal{F}}$, Condition C3 implies

$$\begin{aligned}\left\|\mathcal{T}(Q) - \mathcal{T}^{\mathrm{eval}}_{Q^\star}(Q)\right\|_{2,\mu^\star} &\le \int_0^1 \|(D\mathcal{T}(Q_t) - D\mathcal{T}(Q^\star))[H]\|_{2,\mu^\star}\, dt \\ &\le \omega_{\mathcal{T}}(R)\|H\|_{2,\mu^\star}.\end{aligned}$$

□

D.4 Proof of the local contraction theorem

Proof of Theorem 1. First, since $R \ge \mathrm{dist}(\mathcal{F}, Q^\star)$, $\mathbb{B}_{\mathcal{F}}(R, Q^\star)$ is the intersection of a closed ball and a convex set. It is therefore convex and nonempty because $\Pi_{\mathcal{F}} Q^\star \in \mathcal{F}$ with $\|\Pi_{\mathcal{F}} Q^\star - Q^\star\|_{2,\mu^\star} = \mathrm{dist}(\mathcal{F}, Q^\star) \le R$. Fix $Q_1, Q_2 \in \mathbb{B}_{\mathcal{F}}(R, Q^\star)$ and set $H := Q_1 - Q_2$. For $t \in [0,1]$, define the path $Q_t := Q_2 + tH \in \mathcal{F}$. By the fundamental theorem of calculus in Banach spaces,

$$\mathcal{T}(Q_1) - \mathcal{T}(Q_2) = \int_0^1 D\mathcal{T}(Q_t)[H]\, dt.$$

Taking $L^2(\mu^\star)$ norms and using the triangle inequality,

$$\|\mathcal{T}(Q_1) - \mathcal{T}(Q_2)\|_{2,\mu^\star} \;\le\; \int_0^1 \|D\mathcal{T}(Q_t)[H]\|_{2,\mu^\star}\, dt. \tag{4}$$

We now bound $\|D\mathcal{T}(Q_t)[H]\|_{2,\mu^\star}$ uniformly over $t \in [0,1]$. Since the ball is convex, $Q_t \in \mathbb{B}_{\mathcal{F}}(R, Q^\star)$. Add and subtract the derivative at the optimum:

$$D\mathcal{T}(Q_t)[H] = D\mathcal{T}(Q^\star)[H] + \big(D\mathcal{T}(Q_t) - D\mathcal{T}(Q^\star)\big)[H].$$

Using Lemma 4 and Condition C3, we control each term separately.

First, at $Q^\star$,

$$\|D\mathcal{T}(Q^\star)[H]\|_{2,\mu^\star} = \gamma\, \|P^{\mathrm{eval}}_{\pi^\star} H\|_{2,\mu^\star} \;\le\; \gamma\, \|H\|_{2,\mu^\star}, \tag{5}$$

since $P^{\mathrm{eval}}_{\pi^\star}$ is nonexpansive in $L^2(\mu^\star)$.

Second, because $H \in \mathcal{H}^\star_{\mathcal{F}}$ and $Q_t \in \mathbb{S}^\star_{\mathcal{F}}(R)$,

$$\Big\|\big(D\mathcal{T}(Q_t) - D\mathcal{T}(Q^\star)\big)[H]\Big\|_{2,\mu^\star} \;\le\; \omega_{\mathcal{T}}(R)\, \|H\|_{2,\mu^\star}. \tag{6}$$

Combining the bounds in (5) and (6), we obtain, for all $t \in [0,1]$,

$$\|D\mathcal{T}(Q_t)[H]\|_{2,\mu^\star} \;\le\; \gamma\, \|H\|_{2,\mu^\star} \;+\; \omega_{\mathcal{T}}(R)\, \|H\|_{2,\mu^\star} = \rho_{\mathrm{loc}}(R)\, \|H\|_{2,\mu^\star},$$

where $\rho_{\mathrm{loc}}(R) = \gamma + \omega_{\mathcal{T}}(R)$. Substituting this bound into (4) yields

$$\|\mathcal{T}(Q_1) - \mathcal{T}(Q_2)\|_{2,\mu^\star} \;\le\; \int_0^1 \rho_{\mathrm{loc}}(R)\, \|H\|_{2,\mu^\star}\, dt = \rho_{\mathrm{loc}}(R)\, \|Q_1 - Q_2\|_{2,\mu^\star}.$$

This gives the desired unprojected Lipschitz bound. Since $\Pi_{\mathcal{F}}$ is the metric projection onto the closed convex set $\mathcal{F}$ in $L^2(\mu^\star)$, it is nonexpansive. Therefore

$$\|\mathcal{T}_{\mathcal{F}}(Q_1) - \mathcal{T}_{\mathcal{F}}(Q_2)\|_{2,\mu^\star} \le \|\mathcal{T}(Q_1) - \mathcal{T}(Q_2)\|_{2,\mu^\star} \le \rho_{\mathrm{loc}}(R)\, \|Q_1 - Q_2\|_{2,\mu^\star}.$$

Since $R < r_{\mathrm{loc}}$, $\rho_{\mathrm{loc}}(R) < 1$, so $\mathcal{T}_{\mathcal{F}}$ is a strict contraction on $\mathbb{B}_{\mathcal{F}}(R, Q^\star)$ in the $L^2(\mu^\star)$ norm. □

E Local Fixed Point and Population Convergence

We next collect the fixed-point existence argument and the population convergence proof for the projected soft Bellman iteration.

E.1 Existence of the local fixed point

Banach's fixed-point theorem implies that $\mathcal{T}_{\mathcal{F}}$ admits a locally unique fixed point in a neighborhood of $Q^\star$:

$$Q^\dagger \in \mathcal{F} \quad \text{such that} \quad Q^\dagger = \mathcal{T}_{\mathcal{F}}(Q^\dagger),$$

whenever the gap $\mathrm{dist}(\mathcal{F}, Q^\star)$ is sufficiently small.

Lemma 9 (Existence and local uniqueness of $Q^\dagger$)**.** *Under C1–C3, suppose there exists $R \in (0, r_{\mathrm{loc}})$ such that*

$$\mathrm{dist}(\mathcal{F}, Q^\star) \;\le\; (1 - \rho_{\mathrm{loc}}(R))\, R. \tag{7}$$

Then $\mathcal{T}_{\mathcal{F}}$ maps $\mathbb{B}_{\mathcal{F}}(R, Q^\star)$ into itself and admits a unique fixed point $Q^\dagger$ in this ball.

E.2 Proof of locally unique existence of the fixed point

Proof of Lemma 9. Fix $R \in (0, r_{\mathrm{loc}})$ satisfying (7), and let $Q \in \mathbb{B}_{\mathcal{F}}(R, Q^\star)$. By Theorem 1, the operator $\mathcal{T}_{\mathcal{F}}$ is a $\rho_{\mathrm{loc}}(R)$-contraction on $\mathbb{B}_{\mathcal{F}}(R, Q^\star)$, so in particular

$$\|\mathcal{T}_{\mathcal{F}}(Q) - \mathcal{T}_{\mathcal{F}}(Q^\star)\|_{2,\mu^\star} \;\le\; \rho_{\mathrm{loc}}(R)\, \|Q - Q^\star\|_{2,\mu^\star} \;\le\; \rho_{\mathrm{loc}}(R)\, R.$$

Moreover, since $\mathcal{T}(Q^\star) = Q^\star$, we have $\mathcal{T}_{\mathcal{F}}(Q^\star) = \Pi_{\mathcal{F}} Q^\star$ and hence

$$\|\mathcal{T}_{\mathcal{F}}(Q^\star) - Q^\star\|_{2,\mu^\star} = \|\Pi_{\mathcal{F}} Q^\star - Q^\star\|_{2,\mu^\star} = \mathrm{dist}(\mathcal{F}, Q^\star).$$

By the triangle inequality,

$$\begin{aligned}
&\|\mathcal{T}_{\mathcal{F}}(Q) - Q^\star\|_{2,\mu^\star} \\
&\le \|\mathcal{T}_{\mathcal{F}}(Q) - \mathcal{T}_{\mathcal{F}}(Q^\star)\|_{2,\mu^\star} + \|\mathcal{T}_{\mathcal{F}}(Q^\star) - Q^\star\|_{2,\mu^\star} \\
&\le \rho_{\mathrm{loc}}(R)\, R + \mathrm{dist}(\mathcal{F}, Q^\star) \\
&\le \rho_{\mathrm{loc}}(R)\, R + (1 - \rho_{\mathrm{loc}}(R))\, R \;=\; R,
\end{aligned}$$

where the last inequality uses (7). Thus $\mathcal{T}_{\mathcal{F}}(Q) \in \mathbb{B}_{\mathcal{F}}(R, Q^\star)$, so $\mathcal{T}_{\mathcal{F}}$ maps $\mathbb{B}_{\mathcal{F}}(R, Q^\star)$ into itself.

By Theorem 1, $\mathcal{T}_{\mathcal{F}}$ is a strict contraction on $\mathbb{B}_{\mathcal{F}}(R, Q^\star)$ with respect to $\|\cdot\|_{2,\mu^\star}$. Since $\mathbb{B}_{\mathcal{F}}(R, Q^\star)$ is a closed, convex subset of $L^2(\mu^\star)$ and is nonempty for $R > \mathrm{dist}(\mathcal{F}, Q^\star)$, Banach's fixed-point theorem implies that $\mathcal{T}_{\mathcal{F}}$ admits a unique fixed point $Q^\dagger$ in $\mathbb{B}_{\mathcal{F}}(R, Q^\star)$. ☐

E.3 Proof of local linear convergence

Lemma 10. *Suppose Condition C4 holds. Then there exists $r \in (0, r_{\mathrm{loc}})$ such that*

$$\mathrm{dist}(\mathcal{F}, Q^\star) \;\le\; (1 - \rho_{\mathrm{loc}}(r))\, r,$$

so the assumptions of Lemma 9 are satisfied.

Proof. Condition C4 gives $R_{\mathrm{fp}} \in (0, r_{\mathrm{loc}})$ such that

$$\mathrm{dist}(\mathcal{F}, Q^\star) \le \{1 - \rho_{\mathrm{loc}}(R_{\mathrm{fp}})\} R_{\mathrm{fp}}.$$

Thus the invariance condition of Lemma 9 holds with $r = R_{\mathrm{fp}}$. ☐

Proof of Theorem 2. By Condition C4 and Lemma 10, the projected fixed point $Q^\dagger$ exists.

Step 1: Contraction on a neighborhood of $Q^\dagger$. For any $Q \in \mathbb{B}_\mathcal{F}(r, Q^\dagger)$, the triangle inequality gives

$$\|Q - Q^\star\|_{2,\mu^\star} \;\le\; \|Q - Q^\dagger\|_{2,\mu^\star} + \|Q^\dagger - Q^\star\|_{2,\mu^\star} \;\le\; r + \varepsilon_\mathcal{F}.$$

Thus

$$\mathbb{B}_\mathcal{F}(r, Q^\dagger) \;\subseteq\; \mathbb{B}_\mathcal{F}(r + \varepsilon_\mathcal{F}, Q^\star).$$

Since $\varepsilon_\mathcal{F} \ge \mathrm{dist}(\mathcal{F}, Q^\star)$ and $r + \varepsilon_\mathcal{F} < r_{\mathrm{loc}}$, the local contraction property of Theorem 1 applies with radius $R := r + \varepsilon_\mathcal{F}$ and yields: for all $Q_1, Q_2 \in \mathbb{B}_\mathcal{F}(r, Q^\dagger)$

$$\|\mathcal{T}_\mathcal{F}(Q_1) - \mathcal{T}_\mathcal{F}(Q_2)\|_{2,\mu^\star} \;\le\; \rho_{\mathrm{loc}}(R)\, \|Q_1 - Q_2\|_{2,\mu^\star},$$

where

$$\rho_{\mathrm{loc}}(R) = \rho_{\mathrm{loc}}(r + \varepsilon_\mathcal{F}).$$

Step 2: Invariance and linear convergence. Let $Q^{(0)} \in \mathbb{B}_\mathcal{F}(r, Q^\dagger)$ and define $Q^{(k+1)} = \mathcal{T}_\mathcal{F}(Q^{(k)})$. Since $Q^\dagger = \mathcal{T}_\mathcal{F}(Q^\dagger)$, by Theorem 1, we have for any $Q \in \mathbb{B}_\mathcal{F}(r, Q^\dagger)$,

$$\begin{aligned}\|\mathcal{T}_\mathcal{F}(Q) - Q^\dagger\|_{2,\mu^\star} &= \|\mathcal{T}_\mathcal{F}(Q) - \mathcal{T}_\mathcal{F}(Q^\dagger)\|_{2,\mu^\star} \\ &\le\; \rho_{\mathrm{loc}}(r + \varepsilon_\mathcal{F})\, \|Q - Q^\dagger\|_{2,\mu^\star}.\end{aligned}$$

Since $\rho_{\mathrm{loc}}(r + \varepsilon_\mathcal{F}) < 1$, it follows that $\mathcal{T}_\mathcal{F}(Q) \in \mathbb{B}_\mathcal{F}(r, Q^\dagger)$, so the ball is invariant.

By induction, $Q^{(k)} \in \mathbb{B}_\mathcal{F}(r, Q^\dagger)$ for all $k \ge 0$, and

$$\|Q^{(k)} - Q^\dagger\|_{2,\mu^\star} \;\le\; \{\rho_{\mathrm{loc}}(r + \varepsilon_\mathcal{F})\}^k\, \|Q^{(0)} - Q^\dagger\|_{2,\mu^\star},$$

which proves the claim.

□

E.4 Contraction-rate tightening

Theorem 2 is intentionally conservative: the modulus $\rho_{\mathrm{loc}}(r + \varepsilon_\mathcal{F})$ depends on the initialization radius r and gives a uniform bound over all later iterations. As the iterates approach $Q^\dagger$, they enter smaller basins where contraction is stronger. In particular, if $Q^{(K)} \in \mathbb{B}_\mathcal{F}(r', Q^\dagger)$ with $r' < r$, we may reapply Theorem 2 with initialization $Q^{(0)} := Q^{(K)}$, yielding the sharper modulus $\rho_{\mathrm{loc}}(r' + \varepsilon_\mathcal{F})$. Along the trajectory, the relevant modulus decreases toward $\rho_{\mathrm{loc}}(\varepsilon_\mathcal{F})$. Under misspecification, this is the smallest modulus reachable by centering the local ball at $Q^\dagger$, and it approaches γ only when both $\varepsilon_\mathcal{F}$ and the curvature modulus are small.

The following corollary shows that, in the realizable case, the observed contraction modulus tightens along the trajectory; the familiar geometric envelope is recovered under the appendix interpolation check.

Corollary 2 (Finite-time tightening of the contraction rate). *Assume $Q^\dagger = Q^\star$ and the conditions of Theorem 2. Define $e_k := \|Q^{(k)} - Q^\star\|_{2,\mu^\star}$ and $\rho := \rho_{\mathrm{loc}}(r)$. Then the per-iteration contraction modulus satisfies*

$$\rho_k := \frac{e_{k+1}}{e_k} \le \rho_{\mathrm{loc}}(e_k) = \gamma + \omega_\mathcal{T}(e_k), \qquad k \ge 0 \textit{ whenever } e_k > 0.$$

If, in addition, the interpolation sufficient condition in Lemma 7 holds, then

$$\rho_k - \gamma \le \beta_{\mathrm{loc}} e_0^\alpha \rho^{\alpha k}, \qquad k \ge 0 \textit{ whenever } e_k > 0.$$

Thus the intrinsic modulus $\omega_\mathcal{T}$ controls how quickly the observed contraction improves along the trajectory. Under the L^∞–L^2 interpolation route, the exponent α gives the explicit polynomial envelope; when $\alpha = 1$, the remainder is quadratic and $\rho_k - \gamma \le \beta_{\mathrm{loc}} e_0 \rho^k$.

Proof of Corollary 2. Under the assumption $Q^\dagger = Q^\star$ we have $\varepsilon_\mathcal{F} = 0$, so Theorem 2 yields for any $Q^{(0)} \in \mathbb{B}_\mathcal{F}(r, Q^\star)$ and all $k \ge 0$,

$$e_k = \|Q^{(k)} - Q^\star\|_{2,\mu^\star} \;\le\; \rho^k\, e_0, \quad \rho := \rho_{\mathrm{loc}}(r).$$

If $e_k = 0$, the iterate is already at $Q^\star$ and the claim is trivial. Otherwise, by the local contraction bound applied on the smaller ball $\mathbb{B}_{\mathcal{F}}(e_k, Q^\star)$,

$$e_{k+1} \;\le\; \rho_{\rm loc}(e_k) e_k,$$

so the per-iteration modulus

$$\rho_k := \frac{e_{k+1}}{e_k}$$

satisfies

$$\rho_k \;\le\; \rho_{\rm loc}(e_k) = \gamma + \omega_{\mathcal{T}}(e_k).$$

If Lemma 7 applies, then $\omega_{\mathcal{T}}(u) \le \beta_{\rm loc} u^\alpha$. Using the bound from Theorem 2,

$$e_k^\alpha \;\le\; (\rho^k e_0)^\alpha = e_0^\alpha\, \rho^{\alpha k}.$$

Combining the last two displays gives

$$\rho_k - \gamma \;\le\; \beta_{\rm loc}\, e_0^\alpha\, \rho^{\alpha k}, \qquad k \ge 0,$$

which is the claimed finite-time tightening of the contraction rate. □

F Weighted Regression and Finite-Sample Tools

This appendix gathers the population-risk identity and empirical-process tools used to control one weighted Bellman regression step. Ratios with $d^\star$ in the denominator are interpreted on $\{d^\star > 0\}$; equivalently, the corresponding essential suprema are taken over this support.

F.1 Population-risk identity

Let $\mathcal{F}$ be a convex function class and let $\hat{Q}^{\rm init} \in \mathcal{F}$ be an initial (possibly data–dependent) estimate. Write

$$V_{\hat{Q}^{\rm init}}(s') := \tau \log \sum_{a' \in \mathcal{A}} \exp\{\hat{Q}^{\rm init}(s', a')/\tau\}$$

for the corresponding soft value.

In what follows, expectations are taken with the nuisance estimators held fixed. Define the stationary-weighted population risk

$$\hat{R}_0(Q) := \mathbb{E}_{\nu_b}\Big[d^\star(S,A)\left\{R + \gamma V_{\hat{Q}^{\rm init}}(S') - Q(S,A)\right\}^2\Big].$$

Lemma 11 shows that this stationary-reweighted population objective is correctly specified for the projected soft Bellman operator $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\rm init})$.

Lemma 11 (Fixed-point operator as population risk minimizer)**.**

$$\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\rm init}) = \arg\min_{Q \in \overline{\mathcal{F}}} \hat{R}_0(Q).$$

Proof. Let

$$Y^{\rm init} := R + \gamma V_{\hat{Q}^{\rm init}}(S')$$

and denote its conditional mean by

$$g(S,A) := \mathbb{E}\big[Y^{\rm init} \mid S, A\big] = \mathcal{T}(\hat{Q}^{\rm init})(S,A).$$

Since $d^\star \nu_b = \mu^\star$, $\hat{R}_0(Q)$ equals

$$\mathbb{E}_{\mu^\star}\Big[\left\{Y^{\rm init} - Q(S,A)\right\}^2\Big].$$

By the law of total expectation, this objective differs from $\|\mathcal{T}(\hat{Q}^{\rm init}) - Q\|_{2,\mu^\star}^2$ by a constant that does not depend on Q. Therefore, by definition of the projected Bellman operator,

$$\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\rm init}) = \arg\min_{Q \in \overline{\mathcal{F}}} \hat{R}_0(Q),$$

since this is precisely the $L^2(\mu^\star)$ projection of the soft Bellman target g onto $\overline{\mathcal{F}}$. □

F.2 Empirical-process preliminaries

F.2.1 Local maximal inequality

Let $O_1, \ldots, O_n \in \mathcal{O}$ be independent random variables. For any function $f : \mathcal{O} \to \mathbb{R}$, define

$$\|f\| := \sqrt{\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[f(O_i)^2]}. \tag{8}$$

For a star-shaped class of functions $\mathcal{F}$ and a radius $\delta \in (0, \infty)$, define the localized Rademacher complexity

$$\mathcal{R}_n(\mathcal{F}, \delta) := \mathbb{E}\left[\sup_{\substack{f\in\mathcal{F}\\ \|f\|\le\delta}} \frac{1}{n}\sum_{i=1}^{n}\epsilon_i f(O_i)\right],$$

where ϵ_i are i.i.d. Rademacher random variables.

The following lemma provides a local maximal inequality and restates Lemma 11 of [22]; see also Lemma 11 of van der Laan et al. [73].

Lemma 12 (Local maximal inequality)**.** *Let $\mathcal{F}$ be a star-shaped class of functions satisfying $\sup_{f\in\mathcal{F}} \|f\|_\infty \le M$. Let $\delta = \delta_n \in (0, 1)$ satisfy the critical radius condition $\mathcal{R}_n(\mathcal{F}, \delta) \le \delta^2$, and suppose that, as $n \to \infty$,*

$$\frac{1}{\sqrt{n}}\sqrt{\log\log(1/\delta_n)} = o(\delta_n).$$

Then there exists a universal constant $C > 0$ such that, for all $\eta \in (0, 1)$, with probability at least $1 - \eta$, every $f \in \mathcal{F}$ satisfies

$$\left|\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big)\right| \le C\Big(\delta^2 + \delta\,\|f\|\Big) + C\Big(\frac{\sqrt{\log(e/\eta)}\,\|f\|}{\sqrt{n}} + \frac{M\,\log(e/\eta)}{n}\Big).$$

Proof. Apply the cited one-sided inequality to the symmetric class $\mathcal{F}_\pm := \mathcal{F} \cup (-\mathcal{F})$. This class has the same envelope and the same uniform entropy integral as $\mathcal{F}$, up to universal constants, and remains star-shaped. Lemma 11 in van der Laan et al. [73] shows that there exists a universal constant $C > 0$ such that, for all $u \ge 1$, with probability at least $1 - e^{-u^2}$, every $f \in \mathcal{F}_\pm$ satisfies

$$\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big) \le C\Big(\delta^2 + \delta\,\|f\| + \frac{u\,\|f\|}{\sqrt{n}} + \frac{M\,u^2}{n}\Big).$$

Set $u := \sqrt{\log(e/\eta)}$, so that $e^{-u^2} = e^{-\log(e/\eta)} = \eta/e \le \eta$. Substituting this choice of u into the above inequality yields, with probability at least $1 - \eta$, every $f \in \mathcal{F}$ satisfies

$$\left|\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big)\right| \le C\Big(\delta^2 + \delta\,\|f\|\Big) + C\Big(\frac{\sqrt{\log(e/\eta)}\,\|f\|}{\sqrt{n}} + \frac{M\,\log(e/\eta)}{n}\Big).$$

□

The following lemma bounds the localized Rademacher complexity in terms of the uniform entropy integral and is a direct consequence of Theorem 2.1 of Van Der Vaart and Wellner [74].

For any distribution Q and any uniformly bounded function class $\mathcal{F}$, let $N(\varepsilon, \mathcal{F}, L^2(Q))$ denote the ε-covering number of $\mathcal{F}$ under the $L^2(Q)$ norm [75]. Define the uniform entropy integral of $\mathcal{F}$ by

$$\mathcal{J}(\delta, \mathcal{F}) := \int_0^\delta \sup_Q \sqrt{\log N(\epsilon, \mathcal{F}, L^2(Q))}\, d\epsilon,$$

where the supremum is taken over all discrete probability distributions Q.

Lemma 13. *Let $\mathcal{F}$ be a star-shaped class of functions such that $\sup_{f\in\mathcal{F}} \|f\|_\infty \le M$. Then, for every $\delta > 0$,*

$$\mathcal{R}_n(\mathcal{F},\delta) \;\lesssim\; \frac{1}{\sqrt{n}}\,\mathcal{J}(\delta,\mathcal{F})\left(1+\frac{\mathcal{J}(\delta,\mathcal{F})}{\delta\sqrt{n}}\right),$$

where the implicit constant depends only on M.

Proof. This bound follows directly from the argument in the proof of Theorem 2.1 of Van Der Vaart and Wellner [74]; see in particular the step where the local Rademacher complexity is controlled by the uniform entropy integral for star-shaped classes. □

Lemma 14 (Empirical-process bound for one weighted regression)**.** *Assume Conditions C2, C5, and C8. Let P_0 denote the law of (S,A,R,S') under ν_b and dynamics P, and let P_n be the empirical measure. Let $B_w \ge 1$ and let w be any state–action function satisfying $\|w\|_\infty \le B_w$.*

For each $Q \in \mathcal{F}$, write

$$V_Q(s') := \tau \log \sum_{a'\in\mathcal{A}} \exp\{Q(s',a')/\tau\}.$$

Then there exists a constant $C = C(M,\tau\log|\mathcal{A}|) > 0$ such that, for all $\eta\in(0,1)$, with probability at least $1-\eta$, simultaneously for all $Q_1,Q_2\in\mathcal{F}$ and $V_1 \in \mathcal{V}_\mathcal{F} := \{V_Q : Q\in\mathcal{F}\}$,

$$\begin{aligned}
&\Big|(P_n-P_0)\big[w\,\big(Q_1-Q_2\big)(S,A) \\
&\qquad\qquad \times \big\{R+\gamma V_1(S')-Q_2(S,A)\big\}\big]\Big| \\
&\le\; C\bigg(B_w\delta_n^2+\delta_n\,\Big\|w\big(Q_1-Q_2\big)\Big\|_{L^2(P_0)}\bigg) \\
&\quad +\; C\left(\frac{\sqrt{\log(e/\eta)}\,\Big\|w\big(Q_1-Q_2\big)\Big\|_{L^2(P_0)}}{\sqrt{n}}\right) \\
&\quad +\; C\,B_w\frac{\log(e/\eta)}{n}.
\end{aligned}$$

Proof of Lemma 14. Fix a bounded weight function w with $\|w\|_\infty \le B_w$. Let

$$\mathcal{V}_\mathcal{F} := \{\,V_Q : Q\in\mathcal{F}\,\},\qquad V_Q(s) := \tau\log\sum_{a\in\mathcal{A}}\exp\{Q(s,a)/\tau\}.$$

Define the function class

$$\begin{aligned}\mathcal{G}_w := \Big\{ g_{Q_1,Q_2,V_1} : (s,a,r,s') \mapsto w(s,a)\,\big(Q_1(s,a)-Q_2(s,a)\big) \\ \times\big\{r+\gamma V_1(s')-Q_2(s,a)\big\} :\; Q_1,Q_2\in\mathcal{F},\; V_1\in\mathcal{V}_\mathcal{F}\Big\}.\end{aligned}$$

By Conditions C2 and C5, every $Q\in\mathcal{F}$, every $V_1\in\mathcal{V}_\mathcal{F}$, and R are uniformly bounded by constants depending only on M, and $\tau\log|\mathcal{A}|$. Thus, for any $g_{Q_1,Q_2,V_1}\in\mathcal{G}_w$,

$$\begin{aligned}|g_{Q_1,Q_2,V_1}(s,a,r,s')| &\le |w(s,a)|\,\big|Q_1(s,a)-Q_2(s,a)\big| \\ &\qquad\times\big(|r|+\gamma|V_1(s')|+|Q_2(s,a)|\big) \\ &\le C_0(M,\tau\log|\mathcal{A}|)\,|w(s,a)|\,\big|Q_1(s,a)-Q_2(s,a)\big|.\end{aligned}$$

for some constant $C_0(M,\tau\log|\mathcal{A}|)<\infty$. In particular, $\mathcal{G}_w$ has envelope proportional to B_w, and

$$\|g_{Q_1,Q_2,V_1}\|_{L^2(P_0)} \;\le\; C_0(M,\tau\log|\mathcal{A}|)\,\|w\,(Q_1-Q_2)\|_{L^2(P_0)}. \tag{9}$$

Entropy control. By Condition C8, the uniform entropy integral $\mathcal{J}(\delta, \mathcal{F})$ is finite and determines the critical radius δ_n. For any discrete distribution Q_S on $\mathcal{S}$, let $\bar{Q}$ be the lifted state–action distribution $\bar{Q}(s,a) := Q_S(s)/|\mathcal{A}|$. By Condition C2, for any $Q, \widetilde{Q} \in \mathcal{F}$, the mean-value theorem and the finite-action softmax gradient bound give

$$\left| V_Q(s) - V_{\widetilde{Q}}(s) \right| \le C_{M,\tau \log |\mathcal{A}|} \|Q(s,\cdot) - \widetilde{Q}(s,\cdot)\|_{\ell_2(\mathrm{Unif}(\mathcal{A}))}.$$

Therefore

$$\|V_Q - V_{\widetilde{Q}}\|_{L^2(Q_S)} \le C_{M,\tau \log |\mathcal{A}|} \|Q - \widetilde{Q}\|_{L^2(\bar{Q})}.$$

It follows that, for every $\varepsilon > 0$,

$$N(\varepsilon, \mathcal{V}_{\mathcal{F}}, L^2(Q_S)) \le N(\varepsilon / C_{M,\tau \log |\mathcal{A}|}, \mathcal{F}, L^2(\bar{Q})).$$

Since the uniform entropy supremum for $\mathcal{F}$ ranges over all discrete state–action distributions, the uniform entropy of $\mathcal{V}_{\mathcal{F}}$ is controlled by that of $\mathcal{F}$, up to constants depending only on M and $\tau \log |\mathcal{A}|$, and up to a rescaling of the radius. The map $(Q_1, Q_2, V_1) \mapsto g_{Q_1,Q_2,V_1}$ is a Lipschitz algebraic transformation of (Q_1, Q_2, V_1), with Lipschitz constant depending only on M, and $\tau \log |\mathcal{A}|$, after normalizing the weight by B_w. By standard permanence properties of entropy under Lipschitz maps and bounded multipliers (e.g., Thm. 2.6.18 in [75]), the local entropy of the normalized class $\mathcal{G}_{w/B_w}$ is therefore controlled by that of $\mathcal{F}$: there exists $C_1 = C_1(M, \tau \log |\mathcal{A}|)$ such that

$$\mathcal{J}(\delta, \mathcal{G}_{w/B_w}) \;\le\; C_1\, \mathcal{J}(C_1 \delta, \mathcal{F}), \qquad \forall\, \delta > 0.$$

Hence $\mathcal{G}_{w/B_w}$ has the same critical radius as $\mathcal{F}$, up to a multiplicative constant depending only on M, and $\tau \log |\mathcal{A}|$. We therefore continue to denote this radius by δ_n.

Finally, we replace $\mathcal{G}_{w/B_w}$ by its star-shaped hull

$$\mathcal{G}^{\circ}_{w/B_w} := \{\, tg : g \in \mathcal{G}_{w/B_w},\ t \in [0,1] \,\}.$$

Standard permanence properties of uniform entropy numbers (e.g., Thm. 2.6.18 in [75]) guarantee that $\mathcal{J}(\delta, \mathcal{G}^{\circ}_{w/B_w})$ is bounded, up to a universal constant factor, by $\mathcal{J}(\delta, \mathcal{G}_{w/B_w})$. Hence $\mathcal{G}^{\circ}_{w/B_w}$ has the same critical radius δ_n, up to constants depending only on M, and $\tau \log |\mathcal{A}|$. Because the localized maximal inequality applies to star-shaped classes, we work henceforth with $\mathcal{G}^{\circ}_{w/B_w}$ (without changing any resulting bounds).

Local empirical-process bound. The class $\mathcal{G}^{\circ}_{w/B_w}$ is uniformly bounded, star-shaped, and—by the entropy bound above—has localized entropy controlled by that of $\mathcal{F}$. Lemma 13 therefore yields $\mathcal{R}_n(\mathcal{G}^{\circ}_{w/B_w}, \delta_n) \lesssim \delta_n^2$. Since δ_n satisfies the critical-radius condition, Lemma 12 applies to $\mathcal{G}^{\circ}_{w/B_w}$ and gives: for all $\eta \in (0,1)$, with probability at least $1 - \eta$, every $g \in \mathcal{G}^{\circ}_{w/B_w}$ satisfies

$$\begin{aligned} |(P_n - P_0) g| \;&\le\; C\big(\delta_n^2 + \delta_n \,\|g\|_{L^2(P_0)}\big) \\ &\quad+\; C\left(\frac{\sqrt{\log(e/\eta)}\, \|g\|_{L^2(P_0)}}{\sqrt{n}} \;+\; \frac{\log(e/\eta)}{n} \right), \end{aligned}$$

for some constant $C = C(M, \tau \log |\mathcal{A}|)$. Applying the same display to the normalized weight w/B_w and multiplying by B_w gives the stated dependence on B_w; the terms involving $\|w(Q_1 - Q_2)\|_{L^2(P_0)}$ are unchanged by this rescaling.

Specialization to the inexact regression function. For iteration k, set

$$Q_1 = \mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}), \qquad Q_2 = \widehat{Q}^{(k+1)}, \qquad V_1 = V_{\widehat{Q}^{(k)}},$$

so Q_1, Q_2 lie in the projection class used by the regression update and $V_1 \in \mathcal{V}_{\mathcal{F}}$. The event just constructed is uniform over these classes, so this specialization is valid even when $\widehat{Q}^{(k)}$ and $\widehat{Q}^{(k+1)}$ are fitted from the same regression sample $\mathcal{D}_n$. The associated regression function $g_k(s,a,r,s')$

$$\begin{aligned} &g_k(s,a,r,s') \\ &\quad = w(s,a)\,(Q_1 - Q_2)(s,a)\,\big\{ r + \gamma V_1(s') - Q_2(s,a) \big\} \end{aligned}$$

belongs to $\mathcal{G}_w$. By (9) and boundedness of R, $\mathcal{F}$, and $\mathcal{V}_{\mathcal{F}}$,

$$\|g_k\|_{L^2(P_0)} \;\le\; C_0(M, \tau \log|\mathcal{A}|)\, \big\| w\left(\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\right) \big\|_{L^2(P_0)}.$$

Applying the uniform deviation bound to g_k and absorbing non-coverage constants into $C = C(M, \tau \log|\mathcal{A}|)$ gives, with probability at least $1-\eta$,

$$\begin{aligned}
&\Big| (P_n - P_0)\big[w\left(\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\right) \\
&\qquad\qquad \times \big\{ R + \gamma V_{\widehat{Q}^{(k)}}(S') - \widehat{Q}^{(k+1)}(S,A) \big\} \big] \Big| \\
&\le C\Big(B_w \delta_n^2 + \delta_n \,\big\| w\big(\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\big) \big\|_{L^2(P_0)} \Big) \\
&\quad + C\, \frac{\sqrt{\log(e/\eta)}\, \big\| w\big(\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\big) \big\|_{L^2(P_0)}}{\sqrt{n}} \\
&\quad + C\, B_w \frac{\log(e/\eta)}{n}.
\end{aligned}$$

which is the desired bound. $\square$

F.3 Proof of the one-step regression error bound

Proof of Lemma 3. Fix $0 \le k \le K-1$ and abbreviate

$$w := d^{\star}, \quad \widehat{Q} := \widehat{Q}^{(k)}, \quad \widehat{Q}^{+} := \widehat{Q}^{(k+1)}.$$

Let P_0 denote the law of (S, A, R, S') under ν_b and P_n the empirical measure on $\mathcal{D}_n$.

Step 1: Empirical first-order optimality. By construction, $\widehat{Q}^{+}$ is the (weighted) empirical risk minimizer:

$$\widehat{Q}^{+} = \operatorname*{arg\,min}_{Q \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \widehat{d}^{(k)}(S_i, A_i) \left\{ R_i + \gamma V_{\widehat{Q}}(S_i') - Q(S_i, A_i) \right\}^2.$$

By convexity of $\mathcal{F}$ and first-order optimality, for all $Q \in \mathcal{F}$,

$$\begin{aligned}
&\frac{1}{n} \sum_{i=1}^{n} \widehat{d}^{(k)}(S_i, A_i) \left\{ Q(S_i, A_i) - \widehat{Q}^{+}(S_i, A_i) \right\} \\
&\qquad\qquad \times \left\{ R_i + \gamma V_{\widehat{Q}}(S_i') - \widehat{Q}^{+}(S_i, A_i) \right\} \;\le\; 0.
\end{aligned}$$

By Lemma 11, the projected Bellman update $\mathcal{T}_{\mathcal{F}}(\widehat{Q})$ belongs to $\mathcal{F}$. Taking $Q = \mathcal{T}_{\mathcal{F}}(\widehat{Q})$ and rearranging, we obtain

$$\begin{aligned}
&P_0\big[\widehat{d}^{(k)}\, \Delta_k \left\{ R + \gamma V_{\widehat{Q}}(S') - \widehat{Q}^{+}(S,A) \right\} \big] \\
&\le\; (P_0 - P_n)\big[\widehat{d}^{(k)}\, \Delta_k \left\{ R + \gamma V_{\widehat{Q}}(S') - \widehat{Q}^{+}(S,A) \right\} \big],
\end{aligned} \tag{10}$$

where we set

$$\Delta_k := \mathcal{T}_{\mathcal{F}}(\widehat{Q}) - \widehat{Q}^{+}.$$

Step 2: Population quadratic lower bound. Using the Bellman target

$$\mathcal{T}(\widehat{Q})(S, A) = \mathbb{E}\big[R + \gamma V_{\widehat{Q}}(S') \mid S, A \big],$$

the law of total expectation yields

$$\begin{aligned}
&P_0\big[\widehat{d}^{(k)}\, \Delta_k \left\{ R + \gamma V_{\widehat{Q}}(S') - \widehat{Q}^{+} \right\} \big] \\
&\qquad = P_0\big[\widehat{d}^{(k)}\, \Delta_k \left\{ \mathcal{T}(\widehat{Q}) - \widehat{Q}^{+} \right\} \big].
\end{aligned}$$

Writing

$$\mathcal{T}(\widehat{Q}) - \widehat{Q}^{+} = \{ \mathcal{T}(\widehat{Q}) - \mathcal{T}_{\mathcal{F}}(\widehat{Q}) \} + \Delta_k,$$

and using $\Delta_k \in \mathcal{H}_{\mathcal{F}}$, the weighted-loss curvature condition gives

$$P_0[\widehat{d}^{(k)}\Delta_k^2] \geq (1-\chi_{\mathcal{H},k})\|\Delta_k\|_{2,w\nu_b}^2.$$

Let

$$B_k := \mathcal{T}(\widehat{Q}) - \mathcal{T}_{\mathcal{F}}(\widehat{Q}).$$

Since $\mathcal{T}_{\mathcal{F}}(\widehat{Q})$ is the metric projection of $\mathcal{T}(\widehat{Q})$ onto the closed convex class $\mathcal{F}$, the projection variational inequality with $Q = \widehat{Q}^+$ gives $P_0[w\,\Delta_k B_k] \geq 0$. Therefore, by Cauchy–Schwarz and the definition of $\omega_{\mathrm{Bell},d^\star}(k)$,

$$P_0[(\widehat{d}^{(k)} - w)\Delta_k B_k] \geq -\omega_{\mathrm{Bell},d^\star}(k)\|\Delta_k\|_{2,w\nu_b}.$$

Combining these pieces, we obtain the population lower bound

$$\begin{aligned} &P_0\big[\widehat{d}^{(k)}\,\Delta_k\,\{R + \gamma V_{\widehat{Q}}(S') - \widehat{Q}^+\}\big] \\ &\geq (1-\chi_{\mathcal{H},k})\|\Delta_k\|_{2,w\nu_b}^2 - \omega_{\mathrm{Bell},d^\star}(k)\|\Delta_k\|_{2,w\nu_b}. \end{aligned} \tag{11}$$

Step 3: Combine with empirical-process deviation (behavior norm). Plugging (11) into (10) and rearranging gives

$$\begin{aligned} (1-\chi_{\mathcal{H},k})\|\Delta_k\|_{2,w\nu_b}^2 \leq\ & (P_0 - P_n)\big[\widehat{d}^{(k)}\,\Delta_k\,\{R + \gamma V_{\widehat{Q}}(S') - \widehat{Q}^+\}\big] \\ & + \omega_{\mathrm{Bell},d^\star}(k)\|\Delta_k\|_{2,w\nu_b}. \end{aligned} \tag{12}$$

By Condition C7, condition on the independently estimated weight sequence. Conditional on these weights, the multipliers $d^\star$ and $\widehat{d}^{(k)} - d^\star$ are fixed bounded functions. Lemma 14 applies uniformly over $Q_1, Q_2 \in \mathcal{F}$ and $V_1 \in \mathcal{V}_{\mathcal{F}}$, so it may be specialized to the data-dependent choices $\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)})$, $\widehat{Q}^{(k+1)}$, and $V_{\widehat{Q}^{(k)}}$ even when all regression iterates are fitted on the same $\mathcal{D}_n$. Let $\xi_k := R + \gamma V_{\widehat{Q}}(S') - \widehat{Q}^+(S,A)$. Then

$$(P_0 - P_n)[\widehat{d}^{(k)}\Delta_k\xi_k] = (P_0 - P_n)[d^\star\Delta_k\xi_k] + (P_0 - P_n)[(\widehat{d}^{(k)} - d^\star)\Delta_k\xi_k].$$

Condition C6 gives

$$\|d^\star\Delta_k\|_{L^2(P_0)} \leq \kappa_{\mathrm{cov}}\|\Delta_k\|_{2,w\nu_b}, \qquad \|(\widehat{d}^{(k)} - d^\star)\Delta_k\|_{L^2(P_0)} \leq \kappa_{\mathrm{cov}}\|\Delta_k\|_{2,w\nu_b}.$$

Applying Lemma 14 with confidence level $\eta/(2K)$ to each of the two multipliers and then taking a union bound over $k = 0, \ldots, K-1$ gives: with probability at least $1-\eta$, for all k,

$$\begin{aligned} &\Big|(P_0 - P_n)\big[\widehat{d}^{(k)}\,\Delta_k\,\{R + \gamma V_{\widehat{Q}}(S') - \widehat{Q}^+\}\big]\Big| \\ &\leq C\Big(\kappa_{\mathrm{cov}}\delta_n^2 + \kappa_{\mathrm{cov}}\delta_n\,\|\Delta_k\|_{2,w\nu_b}\Big) + C\Big(\kappa_{\mathrm{cov}}\frac{\sqrt{\log(eK/\eta)}}{\sqrt{n}}\,\|\Delta_k\|_{2,w\nu_b} + \kappa_{\mathrm{cov}}\frac{\log(eK/\eta)}{n}\Big). \end{aligned}$$

By definition of $\delta_{\mathrm{stat}}(n,\eta,K)$,

$$\delta_n^2 + \tfrac{\log(eK/\eta)}{n} \;\lesssim\; \delta_{\mathrm{stat}}(n,\eta,K)^2, \qquad \delta_n + \tfrac{\sqrt{\log(eK/\eta)}}{\sqrt{n}} \;\lesssim\; \delta_{\mathrm{stat}}(n,\eta,K),$$

so

$$\begin{aligned} &\Big|(P_0 - P_n)\big[\widehat{d}^{(k)}\,\Delta_k\,\{R + \gamma V_{\widehat{Q}}(S') - \widehat{Q}^+\}\big]\Big| \\ &\quad\leq C\Big(\kappa_{\mathrm{cov}}\delta_{\mathrm{stat}}(n,\eta,K)^2 + \kappa_{\mathrm{cov}}\delta_{\mathrm{stat}}(n,\eta,K)\,\|\Delta_k\|_{2,w\nu_b}\Big). \end{aligned} \tag{13}$$

Step 4: Solve the scalar inequality. Let

$$x := \|\Delta_k\|_{2,w\nu_b}, \qquad a := \kappa_{\mathrm{cov}}\delta_{\mathrm{stat}}(n,\eta,K), \qquad \omega := \omega_{\mathrm{Bell},d^\star}(k), \qquad \chi := \chi_{\mathcal{H},k}.$$

Combining (12) with the empirical-process bound (13) gives

$$(1-\chi)x^2 \leq C(a^2 + ax) + \omega x.$$

Young's inequality and $\chi < 1$ imply

$$x \leq \frac{C}{1-\chi}\{a + \omega\}.$$

Since $w\nu_b = \mu^\star$, this is the claimed stationary-norm bound.

$\square$

G Main Finite-Sample Convergence Proofs

We now combine the inexact Picard recursion with the one-step regression bound to prove the finite-sample convergence theorems.

G.1 Proof of the inexact Picard iteration lemma

Proof of Lemma 2. Let $\mathcal{T} := \mathcal{T}_{\mathcal{F}}$ and

$$e_k := \|\widehat{Q}^{(k)} - Q^\dagger\|_{2,\mu^\star}, \qquad k \geq 0.$$

By the contraction argument underlying Theorem 2, for any $r > 0$ satisfying $r + \varepsilon_{\mathcal{F}} < r_{\mathrm{loc}}$, the operator $\mathcal{T}$ is a contraction on the ball

$$\mathbb{B}_{\mathcal{F}}(r, Q^\dagger) := \{Q \in \mathcal{F} : \|Q - Q^\dagger\|_{2,\mu^\star} \leq r\}$$

with modulus $\rho_r = \rho_{\mathrm{loc}}(r + \varepsilon_{\mathcal{F}})$, that is,

$$\|\mathcal{T}(Q) - Q^\dagger\|_{2,\mu^\star} \leq \rho_r \|Q - Q^\dagger\|_{2,\mu^\star} \quad \text{for all } Q \in \mathbb{B}_{\mathcal{F}}(r, Q^\dagger).$$

Step 1: Staying inside the basin. We first show by induction that $e_k \leq r$ for all $k \geq 0$, so that $\widehat{Q}^{(k)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\dagger)$ and the local contraction property applies at each iterate.

The base case $k = 0$ holds by assumption: $e_0 = \|\widehat{Q}^{(0)} - Q^\dagger\|_{2,\mu^\star} \leq r$.

Now fix $k \geq 0$ and suppose $e_k \leq r$. Then $\widehat{Q}^{(k)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\dagger)$, so

$$\|\mathcal{T}(\widehat{Q}^{(k)}) - Q^\dagger\|_{2,\mu^\star} \leq \rho_r e_k.$$

By the inexact-update assumption,

$$\|\widehat{Q}^{(k+1)} - \mathcal{T}(\widehat{Q}^{(k)})\|_{2,\mu^\star} \leq \xi_k.$$

Therefore,

$$\begin{aligned} e_{k+1} &= \|\widehat{Q}^{(k+1)} - Q^\dagger\|_{2,\mu^\star} \\ &\leq \|\mathcal{T}(\widehat{Q}^{(k)}) - Q^\dagger\|_{2,\mu^\star} + \|\widehat{Q}^{(k+1)} - \mathcal{T}(\widehat{Q}^{(k)})\|_{2,\mu^\star} \\ &\leq \rho_r e_k + \xi_k \ \leq \ \rho_r r + (1 - \rho_r) r \ = \ r, \end{aligned}$$

where the last step uses $e_k \leq r$ and $\xi_k \leq (1 - \rho_r) r$. Thus $e_{k+1} \leq r$, completing the induction.

Step 2: Unrolling the inexact recursion. The preceding display gives, for every $k \geq 0$,

$$e_{k+1} \leq \rho_r e_k + \xi_k.$$

Iterating this recursion yields, for any $k \geq 1$,

$$e_k \leq \rho_r^k e_0 + \sum_{j=0}^{k-1} \rho_r^{k-1-j} \xi_j.$$

This is the claimed bound, and the case $k = 0$ is trivial. □

G.2 Proof of Theorem 3

Proof of Theorem 3. Work on the high-probability event from Lemma 3, on which the one-step bound holds uniformly for all $0 \leq k \leq K - 1$. Since $\chi_{\mathcal{H},k} \leq \bar{\chi}_{\mathcal{H}}$, define

$$\xi_k := \frac{C}{1 - \bar{\chi}_{\mathcal{H}}} \{\kappa_{\mathrm{cov}} \delta_{\mathrm{stat}}(n, \eta, K) + \omega_{\mathrm{Bell}, d^\star}(k)\}.$$

Then

$$\|\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\|_{2,\mu^\star} \leq \xi_k.$$

By the definition of $\bar{\xi}_K$, $\xi_k \le \bar{\xi}_K \le (1-\rho_K)r$, where $\rho_K = \rho_K(r) = \rho_{\mathrm{loc}}(r+\varepsilon_{\mathcal{F}})$. Lemma 2 therefore applies and ensures both that the iterates remain in $\mathbb{B}_{\mathcal{F}}(r, Q^\dagger)$ and that, for all $1 \le k \le K$,

$$\|\widehat{Q}^{(k)} - Q^\dagger\|_{2,\mu^\star} \le \rho_K^k \|\widehat{Q}^{(0)} - Q^\dagger\|_{2,\mu^\star} + \sum_{j=0}^{k-1} \rho_K^{k-1-j} \xi_j.$$

Substituting the definition of ξ_j and using $\sum_{j=0}^{k-1} \rho_K^{k-1-j} \le (1-\rho_K)^{-1}$ gives

$$\sum_{j=0}^{k-1} \rho_K^{k-1-j} \xi_j \le \frac{C}{(1-\bar{\chi}_{\mathcal{H}})(1-\rho_K)} \left\{ \kappa_{\mathrm{cov}} \delta_{\mathrm{stat}}(n,\eta,K) + \Omega_{\mathrm{Bell},d^\star}(k) \right\},$$

which is the claimed bound. □

G.3 Coupled residual-interaction convergence bound

The main finite-sample theorem assumes an *a priori* sequence of residual-interaction errors. In practice, these errors depend on the iterates: accurate weights are easier to estimate for policies near the local target. The following companion result closes this feedback loop under a polynomial coupling for $\omega_{\mathrm{Bell},d^\star}(k)$ and uniform weighted-loss curvature stability.

C11) Residual-interaction coupling. For any $\eta \in (0,1)$, there exist constants $C' < \infty$ and $\kappa > 0$ such that, with probability at least $1-\eta$, for all $0 \le k \le K-1$,

$$\omega_{\mathrm{Bell},d^\star}(k) \le C' \|\widehat{Q}^{(k)} - Q^\dagger\|_{2,\mu^\star}^{\kappa} + \delta_{\mathrm{wt}}(n,\eta,K), \qquad \chi_{\mathcal{H},k} \le \bar{\chi}_{\mathcal{H}} < 1.$$

For a candidate radius r, define

$$\xi_\star(r) := \frac{C}{1-\bar{\chi}_{\mathcal{H}}} \{\kappa_{\mathrm{cov}} \delta_{\mathrm{stat}}(n,\eta,K) + C' r^\kappa + \delta_{\mathrm{wt}}(n,\eta,K)\}.$$

C12) Coupled-error basin stability. There exists $r > 0$ such that

$$r + \varepsilon_{\mathcal{F}} < r_{\mathrm{loc}}, \qquad \rho_{\mathrm{loc}}(r+\varepsilon_{\mathcal{F}}) < 1, \qquad \xi_\star(r) \le \{1 - \rho_{\mathrm{loc}}(r+\varepsilon_{\mathcal{F}})\} r.$$

Theorem 4 (Local convergence with coupled residual interactions)**.** *Assume Conditions C1–C8, C9, C11, and C12 hold for some admissible r, and suppose $\widehat{Q}^{(0)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\dagger)$. Let $\bar{\rho} := \rho_{\mathrm{loc}}(r+\varepsilon_{\mathcal{F}})$. Then, with probability at least $1-\eta$, for all $1 \le k \le K$,*

$$\begin{aligned}\|\widehat{Q}^{(k)} - Q^\dagger\|_{2,\mu^\star} \le{}& \bar{\rho}^k \|\widehat{Q}^{(0)} - Q^\dagger\|_{2,\mu^\star} \\ &+ \frac{C}{1-\bar{\rho}} \frac{\kappa_{\mathrm{cov}} \delta_{\mathrm{stat}}(n,\eta,K) + \delta_{\mathrm{wt}}(n,\eta,K)}{1-\bar{\chi}_{\mathcal{H}}} \\ &+ \frac{CC'}{1-\bar{\chi}_{\mathcal{H}}} \sum_{j=0}^{k-1} \bar{\rho}^{k-1-j} e_j^\kappa,\end{aligned}$$

where $e_j := \|\widehat{Q}^{(j)} - Q^\dagger\|_{2,\mu^\star}$.

Proof of Theorem 4. On the event in Condition C11, Lemma 3 gives

$$\|\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\|_{2,\mu^\star} \le \frac{C}{1-\bar{\chi}_{\mathcal{H}}} \{\kappa_{\mathrm{cov}} \delta_{\mathrm{stat}}(n,\eta,K) + C' e_k^\kappa + \delta_{\mathrm{wt}}(n,\eta,K)\}.$$

The basin-stability condition implies that this one-step error is at most $(1-\bar{\rho})r$ whenever $e_k \le r$, so the same induction as in Lemma 2 keeps all iterates in the local ball. Unrolling the resulting recursion yields the displayed bound. □

H Supporting Examples and Extensions

The remaining appendices provide examples, ratio-estimation background, annealing heuristics, and related work.

H.1 Extension to admissible weighting functions

For clarity, the main text focuses on $d^\star = d\mu^\star/d\nu_b$. The more general analysis separates projection equivalence from norm comparison.

C13) Weighted projection equivalence. There exists a nonnegative $d \in L^2(\nu_b)$ such that, for all $Q \in \mathcal{F}$,

$$\Pi_{\mathcal{F}}\mathcal{T}Q \in \arg\min_{f\in\mathcal{F}} \mathbb{E}_{\nu_b}\left[d(S,A)\{\mathcal{T}Q(S,A) - f(S,A)\}^2\right].$$

Exact stationary ratios are the canonical admissible weights when $\mu^\star \ll \nu_b$, but other weights may also be admissible if they induce the same projected Bellman update on $\mathcal{F}$.

C14) Weighted–stationary norm comparison. There exists $1 \le c < \infty$ such that, for all $f \in L^2(\mu^\star)$,

$$c^{-1}\|f\|_{2,\mu^\star} \le \|f\|_{2,d\nu_b} \le c\|f\|_{2,\mu^\star}.$$

For $d = d^\star$, $d^\star\nu_b = \mu^\star$, so Condition C13 holds by identity of the two projection problems and Condition C14 holds with $c = 1$.

H.2 Sufficient conditions for curvature and weight stability

Condition S1 holds for many standard function classes used in empirical risk minimization. For finite-dimensional linear models of dimension p, it holds with $C_\infty \asymp \sqrt{p}$ and $\alpha = 1$. For reproducing kernel Hilbert spaces with eigenvalue decay $\lambda_j \asymp j^{-2r}$, one may take $\alpha = 2r/(2r+1)$ [45, Lemma 5.1]. It also holds for signed convex hulls of suitable bases [70, Lemma 2], and for d-variate Hölder and Sobolev classes of order $s > d/2$ on bounded domains, where $\alpha = 1 - d/(2s)$ [12, Lemma 4]; see also [1, 65].

Sufficient conditions for weighted-loss stability. Condition C9 is implied by the pointwise relative bound

$$\left\|\frac{\widehat{d}^{(k)}}{d^\star} - 1\right\|_\infty < 1,$$

with ratios interpreted as in Appendix F. More generally,

$$\chi_{\mathcal{H},k} \le B_{\mathcal{H}}\left\|\frac{\widehat{d}^{(k)}}{d^\star} - 1\right\|_{2,d^\star\nu_b}, \qquad B_{\mathcal{H}} := \sup_{h\in\mathcal{H}_{\mathcal{F}}} \frac{\|h\|^2_{4,d^\star\nu_b}}{\|h\|^2_{2,d^\star\nu_b}}.$$

Thus relative $L^2(d^\star\nu_b)$ consistency of the weights is enough whenever the fitted directions are not too spiky. The residual-interaction term admits the simple sufficient bound

$$\omega_{\mathrm{Bell},d^\star}(k) \le \left\|\frac{\widehat{d}^{(k)}}{d^\star} - 1\right\|_\infty \|\mathcal{T}\widehat{Q}^{(k)} - \mathcal{T}_{\mathcal{F}}\widehat{Q}^{(k)}\|_{2,d^\star\nu_b},$$

and sharper interpolation bounds can replace the sup-norm factor when the Bellman residual class has additional smoothness.

H.3 Estimation of stationary ratios

Stationary-weighted soft FQI requires weights approximating the stationary density ratio $d_\pi = \mathrm{d}\mu_\pi/\mathrm{d}\nu_b$. Estimating such ratios is a well-studied problem, often formulated through DICE-style saddle-point objectives [49, 88, 39] or related minimax balancing-weight methods [67, 79]. Stationary ratios are the undiscounted analogues of discounted occupancy ratios and arise as a $\gamma \to 1$ limit under suitable normalization and standard ergodicity conditions.

The identifying condition is stationarity. Since $d_\pi\nu_b = \mu_\pi$ is stationary for (π, P), the ratio satisfies

$$E[d_\pi(S,A)\{g(S,A) - g(S',A')\}] = 0 \qquad \text{for all } g, \tag{14}$$

where $(S, A) \sim \nu_b$, $S' \sim P(\cdot \mid S, A)$, and $A' \sim \pi(\cdot \mid S')$. Conversely, if a nonnegative function d satisfies (14) for all bounded measurable g and $\mathbb{E}_{\nu_b} d(S, A) = 1$, then $d\nu_b$ is a stationary state-action distribution for (π, P). Hence, when the stationary distribution is unique, $d = d_\pi$ ν_b-almost surely.

Minimax and DICE-style estimators enforce (14) over a critic class $\mathcal{G}$ while searching over a ratio class $\mathcal{D}$. One representative empirical estimator uses the exact finite-action conditional expectation

$$\bar{g}_\pi(s') := \sum_{a' \in \mathcal{A}} \pi(a' \mid s') g(s', a')$$

and is

$$\begin{aligned} \widehat{d}_\pi \in \arg\min_{d \in \mathcal{D}} \sup_{g \in \mathcal{G}} &\left\{ \frac{1}{n} \sum_{i=1}^{n} d(S_i, A_i)\{g(S_i, A_i) - \bar{g}_\pi(S_i')\} - \frac{1}{2n} \sum_{i=1}^{n} g(S_i, A_i)^2 \right\} \\ &+ \left(\frac{1}{n} \sum_{i=1}^{n} d(S_i, A_i) - 1 \right)^2, \end{aligned} \tag{15}$$

which is tractable because the action space is finite. Drawing a next action from $\pi(\cdot \mid S_i')$ gives a Monte Carlo approximation to the same moment. The final term enforces normalization. The quadratic penalty in g makes the inner problem strongly concave and turns the supremum into a squared measure of stationarity-moment violation. Other DICE variants use different convex regularizers, discounted-flow constraints, or positivity penalties, but rely on the same stationarity-moment structure.

When $\mathcal{G}$ is linear or an RKHS, the inner maximization often has a closed form. For example, let $g_\theta(x) = \theta^\top \phi(x)$, with $x = (s, a)$, and write $\bar{\phi}_\pi(s') := \sum_{a' \in \mathcal{A}} \pi(a' \mid s') \phi(s', a')$. For fixed d, define

$$\widehat{m}_d := \frac{1}{n} \sum_{i=1}^{n} d(S_i, A_i)\{\phi(S_i, A_i) - \bar{\phi}_\pi(S_i')\}, \qquad \widehat{\Sigma} := \frac{1}{n} \sum_{i=1}^{n} \phi(S_i, A_i)\phi(S_i, A_i)^\top.$$

With ridge regularization $\lambda \|\theta\|_2^2/2$, the inner supremum is

$$\sup_\theta \left\{ \theta^\top \widehat{m}_d - \frac{1}{2} \theta^\top (\widehat{\Sigma} + \lambda I)\theta \right\} = \frac{1}{2} \widehat{m}_d^\top (\widehat{\Sigma} + \lambda I)^{-1} \widehat{m}_d.$$

Thus the estimator chooses d to balance stationary-flow moments in the feature space ϕ. RKHS critics give an analogous kernelized discrepancy between the weighted current state-action distribution and the induced next state-action distribution. These structured critic classes yield stable, computationally simple weights while approximating the stationary projection norm [17, 67, 79, 53].

For numerical stability, implementations often regularize or truncate the ratio. Common choices include enforcing $d \geq 0$, adding an ℓ_2 or entropy penalty on d, clipping large weights, or estimating a discounted stationary ratio with discount close to one. Such regularization trades small bias for lower variance and better conditioning, which is important under limited overlap.

Instantiating the weight-error term. The abstract term $\|\widehat{d}_\pi/d_\pi - 1\|_{2, d_\pi \nu_b}$ can be controlled using guarantees for minimax or saddle-point ratio estimators, together with the overlap conditions used in our main theorem. Existing analyses for conditional moment problems give bounds of this form [17, 9, 53], and finite-sample analyses for minimax off-policy weight estimation are given by Uehara et al. [67, 68], Wang et al. [79]. At a high level, these bounds decompose into ratio-class approximation, critic-class identification error, and statistical error:

$$\inf_{d \in \mathcal{D}} \|d - d_\pi\| + \eta_{\mathcal{G}} + \mathrm{stat}(\mathcal{D}, \mathcal{G}, n),$$

where d_π is the true stationary density ratio. Equivalently, if $d^\dagger \in \arg\min_{d \in \mathcal{D}} \|d - d_\pi\|$, the first term is $\|d^\dagger - d_\pi\|$. The term $\eta_{\mathcal{G}}$ measures the residual stationarity violation not detected by the critic class, and $\mathrm{stat}(\mathcal{D}, \mathcal{G}, n)$ is controlled by the complexity of the two classes. When an estimator is analyzed in the behavior norm, the overlap lower bound translates it to the weighted relative-error norm through

$$\left\| \frac{\widehat{d}_\pi}{d_\pi} - 1 \right\|_{2, d_\pi \nu_b} \leq \underline{d}^{-1/2} \|\widehat{d}_\pi - d_\pi\|_{2, \nu_b}.$$

H.4 Action-Gap Refinement and Population Temperature Annealing

This appendix continues the local theory of Section 3 in the low-temperature regime. The baseline local contraction radius is controlled by $\beta_{\mathrm{loc}}(\tau) = O(1/\tau)$, so the general-purpose radius can shrink as $\tau \downarrow 0$. Under an action gap, the softmax curvature is much smaller near the hard-optimal value function. The results below make this refinement explicit and then state a deterministic population annealing consequence. They do not assert a finite-sample annealing guarantee.

H.4.1 Action-gap refinement of the local theory

For this subsection we write $\mathcal{T}_\tau$ for the soft Bellman operator at temperature τ. To avoid the degeneracy of the soft stationary norm as $\tau \downarrow 0$, the refinement is stated in a fixed reference geometry. Let

$$\bar{\mu}(ds, da) := d^\star_{\mathrm{hard}}(ds)\, \nu(a), \qquad \nu(a) \geq \nu_{\min} > 0,$$

where $d^\star_{\mathrm{hard}}$ is the hard-optimal state marginal and the default choice is the uniform action distribution $\nu(a) = 1/|\mathcal{A}|$. Let $\bar{\Pi}_{\mathcal{F}}$ denote metric projection onto $\mathcal{F}$ in $L^2(\bar{\mu})$ and define the fixed-reference projected population update

$$\bar{\mathcal{T}}_{\mathcal{F},\tau} := \bar{\Pi}_{\mathcal{F}} \mathcal{T}_\tau.$$

Let $Q^\star_\tau$ denote the soft-optimal fixed point and let $\bar{Q}^\dagger_\tau$ denote a fixed point of $\bar{\mathcal{T}}_{\mathcal{F},\tau}$, when it exists. The corresponding fixed-reference approximation error is

$$\bar{\varepsilon}_{\mathcal{F}}(\tau) := \|\bar{Q}^\dagger_\tau - Q^\star_\tau\|_{2,\bar{\mu}}.$$

For $r > 0$ and $Q_0 \in L^2(\bar{\mu})$, write

$$\bar{\mathbb{B}}_{\mathcal{F}}(r, Q_0) := \{Q \in \mathcal{F} : \|Q - Q_0\|_{2,\bar{\mu}} \leq r\}.$$

Finally, define the fixed-reference analogue of the main difference class by

$$\bar{\mathcal{H}}^\star_{\mathcal{F},\tau} := \{Q_1 - Q_2 : Q_1, Q_2 \in \mathcal{F} \cup \mathcal{T}_\tau(\mathcal{F}) \cup \{Q^\star_\tau, \bar{Q}^\dagger_\tau\}\}.$$

Let $\mathcal{S}_{\bar{P}}$ denote a full-measure set for next states S' when $(S, A) \sim \bar{\mu}$ and $S' \sim P(\cdot \mid S, A)$. The gap assumptions below are imposed on this transition-relevant state set.

We use the following structural assumptions only in this appendix.

S1 **Uniform action gap.** There exists $\Delta > 0$ such that, for all $S \in \mathcal{S}_{\bar{P}}$ and all $a \neq A^\star(S)$,

$$Q^\star_{\mathrm{hard}}(S, A^\star(S)) \;\geq\; Q^\star_{\mathrm{hard}}(S, a) + 2\Delta,$$

for a measurable optimal action $A^\star(S)$.

S2 **Gap-stable neighborhood.** There exist $r_{\mathrm{gap}} > 0$ and a convex ambient neighborhood $\mathbb{B}_{\mathrm{gap}}$ of $Q^\star_{\mathrm{hard}}$ such that every $Q \in \mathbb{B}_{\mathrm{gap}}$ satisfies, for all $S \in \mathcal{S}_{\bar{P}}$,

$$Q(S, A^\star(S)) \;\geq\; Q(S, a) + \Delta \qquad \text{for all } a \neq A^\star(S).$$

The radius r_{gap} records the largest $L^2(\bar{\mu})$ radius used below for which the relevant local balls and line segments are assumed to lie in $\mathbb{B}_{\mathrm{gap}}$. A sufficient way to verify this condition is to combine the hard gap in Assumption S1 with a uniform L^∞ control ensuring $\|Q - Q^\star_{\mathrm{hard}}\|_\infty \leq \Delta/2$ on the local region under consideration. Thus gap stability is an assumption used to prove the improved curvature and linearization bounds, rather than a consequence of those bounds.

S3 **Fixed-reference local norm control.** There exist $C_{\bar{\infty}} < \infty$ and $\alpha \in (0, 1]$ such that, for every temperature considered and every $H \in \bar{\mathcal{H}}^\star_{\mathcal{F},\tau}$,

$$\|H\|_\infty \;\leq\; C_{\bar{\infty}} \|H\|^\alpha_{2,\bar{\mu}}.$$

S4 **Fixed-reference Bellman stability.** For every temperature considered, there are constants $\kappa_\tau < \infty$ and $C_P < \infty$ such that $\gamma\kappa_\tau < 1$ and, for all $H \in L^2(\bar{\mu})$,

$$\|P^{\mathrm{eval}}_{\pi^\star_\tau} H\|_{2,\bar{\mu}} \leq \kappa_\tau \|H\|_{2,\bar{\mu}},$$

and

$$\|\mathbb{E}[\|H(S', \cdot)\|_2 \mid S, A]\|_{2,\bar{\mu}} \leq C_P \|H\|_{2,\bar{\mu}}.$$

The first display controls the frozen-policy linearization; the second is the transition/concentrability condition used only in the curvature bound.

Lemma 15 (Local second-derivative bound under an action gap)**.** *Assume the structural assumptions S1–S4. Then there exists a finite constant*

$$\bar{\beta}^{\mathrm{gap}}_{\mathrm{loc}}(\tau) := \frac{\gamma}{\tau}\, C_{\mathrm{ref}}\, C_{\mathrm{gap}}\, e^{-\Delta/\tau}$$

where $C_{\mathrm{ref}} := \sqrt{|\mathcal{A}|}\, C_{\bar{\infty}} C_P$ *and* $C_{\mathrm{gap}} < \infty$*, such that, for all* $Q \in \mathbb{B}_{\mathrm{gap}}$ *and all directions* $H_1 \in \bar{\mathcal{H}}^{\star}_{\mathcal{F},\tau}$*,* $H_2 \in L^2(\bar{\mu})$*,*

$$\left\| D^2 \mathcal{T}_\tau(Q)[H_1, H_2] \right\|_{2,\bar{\mu}} \;\le\; \bar{\beta}^{\mathrm{gap}}_{\mathrm{loc}}(\tau)\, \|H_1\|^{\alpha}_{2,\bar{\mu}}\, \|H_2\|_{2,\bar{\mu}}. \tag{16}$$

In particular, for fixed constants in Assumptions S2–S4, the local curvature decays as $\bar{\beta}^{\mathrm{gap}}_{\mathrm{loc}}(\tau) = O\big(\frac{1}{\tau} e^{-\Delta/\tau}\big)$ *as* $\tau \downarrow 0$*.*

Proof. The proof follows the same structure as Lemma 6, with an improved bound on the covariance term under the action-gap condition. By Lemma 5, $D^2\mathcal{T}_\tau(Q)[H_1, H_2]$ can be written in terms of $\mathrm{Cov}_{\pi_Q}\big(H_1(S', \cdot), H_2(S', \cdot)\big)$. Fix S' and write $h_i := H_i(S', \cdot) \in \mathbb{R}^{|\mathcal{A}|}$ and $p_Q := \pi_Q(\cdot \mid S')$. Since $Q \in \mathbb{B}_{\mathrm{gap}}$, Assumption S2 gives

$$Q(S', A^\star(S')) \;\ge\; Q(S', a) + \Delta \quad \text{for all } a \ne A^\star(S').$$

By the definition of the softmax policy,

$$p_Q(a) = \frac{\exp\{Q(S', a)/\tau\}}{\sum_{b \in \mathcal{A}} \exp\{Q(S', b)/\tau\}}, \qquad a \in \mathcal{A}.$$

For any $a \ne A^\star(S')$ we have

$$\frac{p_Q(a)}{p_Q(A^\star(S'))} = \exp\!\left(\frac{Q(S', a) - Q(S', A^\star(S'))}{\tau} \right) \;\le\; e^{-\Delta/\tau},$$

so $p_Q(a) \le e^{-\Delta/\tau}$ and

$$\sum_{a \ne A^\star(S')} p_Q(a) \;\le\; (|\mathcal{A}| - 1) e^{-\Delta/\tau} =: \varepsilon.$$

Thus $p_Q(A^\star(S')) \ge 1 - \varepsilon$, and we may take $C'_{\mathrm{gap}} := |\mathcal{A}| - 1$ so that $p_Q(A^\star(S')) \ge 1 - C'_{\mathrm{gap}} e^{-\Delta/\tau}$.

Next, consider the covariance matrix $\Sigma_Q := \mathrm{diag}(p_Q) - p_Q p_Q^\top$. For any $h \in \mathbb{R}^{|\mathcal{A}|}$,

$$h^\top \Sigma_Q h = \mathrm{Var}_{\pi_Q}\big(h(A)\big).$$

Let $a^\star := A^\star(S')$, write $\varepsilon := \sum_{a \ne a^\star} p_Q(a)$ as above, and set $m := \mathbb{E}_{\pi_Q}[h(A)]$. Then

$$\begin{aligned}
\mathrm{Var}_{\pi_Q}(h) &= \sum_a p_Q(a) \big(h(a) - m\big)^2 \\
&\le 2 \sum_a p_Q(a) \big(h(a) - h(a^\star)\big)^2 + 2\big(h(a^\star) - m\big)^2 \\
&\le 4 \sum_{a \ne a^\star} p_Q(a) \big(h(a) - h(a^\star)\big)^2 \\
&\le C_{\mathrm{gap}}\, \varepsilon\, \|h\|_2^2,
\end{aligned}$$

where numerical constants are absorbed into C_{gap}. Since $\varepsilon \le C'_{\mathrm{gap}} e^{-\Delta/\tau}$, we obtain

$$\mathrm{Var}_{\pi_Q}(h) \;\le\; C_{\mathrm{gap}} e^{-\Delta/\tau}\, \|h\|_2^2.$$

Applying Cauchy–Schwarz to the covariance,

$$\begin{aligned}
\big|\mathrm{Cov}_{\pi_Q}(h_1, h_2)\big| &\le \sqrt{\mathrm{Var}_{\pi_Q}(h_1)\ \mathrm{Var}_{\pi_Q}(h_2)} \\
&\le C_{\mathrm{gap}} e^{-\Delta/\tau}\, \|h_1\|_2\, \|h_2\|_2.
\end{aligned}$$

For each (s,a), this covariance estimate gives

$$\left|D^2\mathcal{T}_\tau(Q)[H_1,H_2](s,a)\right| \le \frac{\gamma}{\tau} C_{\mathrm{gap}} e^{-\Delta/\tau}\mathbb{E}[\|H_1(S',\cdot)\|_2\,\|H_2(S',\cdot)\|_2 \mid s,a]\,.$$

Since

$$\|H_1(S',\cdot)\|_2 \le \sqrt{|\mathcal{A}|}\,\|H_1\|_\infty \le \sqrt{|\mathcal{A}|}\,C_{\bar\infty}\|H_1\|_{2,\bar\mu}^{\alpha},$$

Assumption S4 implies

$$\begin{aligned}
&\left\|D^2\mathcal{T}_\tau(Q)[H_1,H_2]\right\|_{2,\bar\mu}\\
&\quad\le \frac{\gamma}{\tau} C_{\mathrm{gap}} e^{-\Delta/\tau}\sqrt{|\mathcal{A}|}\,C_{\bar\infty}\|H_1\|_{2,\bar\mu}^{\alpha}\,\|\mathbb{E}[\|H_2(S',\cdot)\|_2 \mid S,A]\|_{2,\bar\mu}\\
&\quad\le \frac{\gamma}{\tau}\sqrt{|\mathcal{A}|}\,C_{\bar\infty}C_P C_{\mathrm{gap}} e^{-\Delta/\tau}\|H_1\|_{2,\bar\mu}^{\alpha}\|H_2\|_{2,\bar\mu}.
\end{aligned}$$

This is (16) with $C_{\mathrm{ref}} = \sqrt{|\mathcal{A}|}\,C_{\bar\infty}C_P$. The bound is uniform over $Q \in \mathbb{B}_{\mathrm{gap}}$ and has no dependence on the soft-policy minimum action probability. ☐

Let

$$\bar r_{\mathrm{loc}}^{\mathrm{gap}}(\tau) := \min\left\{r_{\mathrm{gap}},\ \left(\frac{1-\gamma\kappa_\tau}{\bar\beta_{\mathrm{loc}}^{\mathrm{gap}}(\tau)}\right)^{1/\alpha}\right\}, \tag{17}$$

and, for $R < \bar r_{\mathrm{loc}}^{\mathrm{gap}}(\tau)$,

$$\bar\rho_{\mathrm{loc}}^{\mathrm{gap}}(R,\tau) := \gamma\kappa_\tau + \bar\beta_{\mathrm{loc}}^{\mathrm{gap}}(\tau)R^{\alpha}. \tag{18}$$

Lemma 16 (Action-gap linearization remainder)**.** *Assume Assumptions S1–S4. Let $\mathcal{T}_{\tau,\star}^{\mathrm{eval}}$ denote the linear policy-evaluation operator obtained by freezing the soft-optimal policy at $Q_\tau^\star$. If the line segment joining $Q_\tau^\star$ and $Q \in \mathcal{F}$ is contained in $\mathbb{B}_{\mathrm{gap}}$, then*

$$\left\|\mathcal{T}_\tau(Q) - \mathcal{T}_{\tau,\star}^{\mathrm{eval}}(Q)\right\|_{2,\bar\mu} \ \le\ \frac{\bar\beta_{\mathrm{loc}}^{\mathrm{gap}}(\tau)}{2}\,\|Q - Q_\tau^\star\|_{2,\bar\mu}^{1+\alpha}.$$

Proof. This is the second-order Taylor remainder used in Lemma 1, with Lemma 15 replacing Lemma 6 along the line segment. The first-order term is exactly the frozen-policy evaluation operator at $Q_\tau^\star$, and integrating the second derivative over the segment gives the factor $1/2$. ☐

Theorem 5 (Gap-improved projected local contraction)**.** *Assume Assumptions S1–S4. Fix $\tau > 0$ and suppose $\mathcal{F}$ is closed and convex. Let $R < \bar r_{\mathrm{loc}}^{\mathrm{gap}}(\tau)$, and assume the local ball $\bar{\mathbb{B}}_{\mathcal{F}}(R, Q_\tau^\star)$ and the point $Q_\tau^\star$ are contained in $\mathbb{B}_{\mathrm{gap}}$. Then $\bar{\mathcal{T}}_{\mathcal{F},\tau}$ is a contraction on $\bar{\mathbb{B}}_{\mathcal{F}}(R, Q_\tau^\star)$ with modulus*

$$\bar\rho_{\mathrm{loc}}^{\mathrm{gap}}(R,\tau)\ <\ 1.$$

Proof. The proof is the proof of Theorem 1 with the stationary norm replaced by the fixed reference norm $\|\cdot\|_{2,\bar\mu}$, the linearized contraction constant γ replaced by $\gamma\kappa_\tau$, and the second-derivative bound in Lemma 6 replaced by the gap-improved bound in Lemma 15. The assumption that $Q_\tau^\star$ and $\bar{\mathbb{B}}_{\mathcal{F}}(R, Q_\tau^\star)$ lie in the convex set $\mathbb{B}_{\mathrm{gap}}$ ensures that the improved curvature bound is valid on the whole segment used in the Taylor expansion. The final projected claim follows from nonexpansiveness of $\bar\Pi_{\mathcal{F}}$ in $L^2(\bar\mu)$. ☐

Theorem 6 (Gap-improved local linear convergence)**.** *Assume the conditions of Theorem 5. Suppose $\bar Q_\tau^\dagger$ exists, and $\bar\varepsilon_{\mathcal{F}}(\tau) < \bar r_{\mathrm{loc}}^{\mathrm{gap}}(\tau)$. Let $r > 0$ satisfy*

$$r + \bar\varepsilon_{\mathcal{F}}(\tau) < \bar r_{\mathrm{loc}}^{\mathrm{gap}}(\tau),$$

and assume

$$\{Q_\tau^\star\} \cup \bar{\mathbb{B}}_{\mathcal{F}}\big(r + \bar\varepsilon_{\mathcal{F}}(\tau), Q_\tau^\star\big) \subseteq \mathbb{B}_{\mathrm{gap}}.$$

If $Q^{(0)} \in \bar{\mathbb{B}}_{\mathcal{F}}(r, \bar Q_\tau^\dagger)$ and $Q^{(k+1)} = \bar{\mathcal{T}}_{\mathcal{F},\tau}(Q^{(k)})$, then

$$\|Q^{(k)} - \bar Q_\tau^\dagger\|_{2,\bar\mu}\ \le\ \left\{\bar\rho_{\mathrm{loc}}^{\mathrm{gap}}(r + \bar\varepsilon_{\mathcal{F}}(\tau),\tau)\right\}^k\|Q^{(0)} - \bar Q_\tau^\dagger\|_{2,\bar\mu},$$

where $\bar\rho_{\mathrm{loc}}^{\mathrm{gap}}(r + \bar\varepsilon_{\mathcal{F}}(\tau),\tau) < 1$.

Proof. For any Q in the stated ball,

$$\|Q - Q_\tau^\star\|_{2,\bar\mu} \le r + \bar\varepsilon_{\mathcal{F}}(\tau) < \bar r_{\mathrm{loc}}^{\mathrm{gap}}(\tau).$$

Thus Theorem 5 applies with radius $r + \bar\varepsilon_{\mathcal{F}}(\tau)$, yielding the contraction modulus in (18). The induction and linear-rate conclusion are identical to the proof of Theorem 2. ☐

H.4.2 Population temperature annealing

The preceding fixed-temperature result can be composed along a finite temperature grid if adjacent projected fixed points are close enough that each warm start enters the next basin. The statement below is deliberately deterministic, population-level, and stated entirely in the fixed reference norm $\|\cdot\|_{2,\bar{\mu}}$.

Theorem 7 (Population soft FQI under a finite temperature schedule)**.** *Let $\tau_0 > \tau_1 > \cdots > \tau_J > 0$ be a finite decreasing temperature grid. For each $j = 0, \ldots, J$, suppose the projected fixed point $\bar{Q}^{\dagger}_{\tau_j}$ exists. Assume there is a stage radius $s_j > 0$ satisfying*

$$s_j + \bar{\varepsilon}_{\mathcal{F}}(\tau_j) < \bar{r}^{\mathrm{gap}}_{\mathrm{loc}}(\tau_j)$$

such that, on $\bar{\mathbb{B}}_{\mathcal{F}}(s_j, \bar{Q}^{\dagger}_{\tau_j})$, the exact population update $\bar{\mathcal{T}}_{\mathcal{F},\tau_j}$ is locally contractive toward $\bar{Q}^{\dagger}_{\tau_j}$:

$$\|\bar{\mathcal{T}}_{\mathcal{F},\tau_j}(Q) - \bar{Q}^{\dagger}_{\tau_j}\|_{2,\bar{\mu}} \le \rho_j \, \|Q - \bar{Q}^{\dagger}_{\tau_j}\|_{2,\bar{\mu}}, \qquad Q \in \bar{\mathbb{B}}_{\mathcal{F}}(s_j, \bar{Q}^{\dagger}_{\tau_j}),$$

for some $\rho_j < 1$. By Theorem 6, one may take $\rho_j = \bar{\rho}^{\mathrm{gap}}_{\mathrm{loc}}(s_j + \bar{\varepsilon}_{\mathcal{F}}(\tau_j), \tau_j)$ after verifying the required gap-stable balls at temperature τ_j. Choose tolerances $\eta_j \in (0, s_j)$ and suppose the initial point satisfies

$$\|Q_{0,0} - \bar{Q}^{\dagger}_{\tau_0}\|_{2,\bar{\mu}} \le s_0.$$

Assume the basin-overlap condition

$$\eta_j + \|\bar{Q}^{\dagger}_{\tau_j} - \bar{Q}^{\dagger}_{\tau_{j+1}}\|_{2,\bar{\mu}} \ \le\ s_{j+1}, \qquad j = 0, \ldots, J-1. \tag{19}$$

At stage j, run exact population projected soft FQI at temperature τ_j,

$$Q_{j,m+1} = \bar{\mathcal{T}}_{\mathcal{F},\tau_j}(Q_{j,m}),$$

warm-starting with $Q_{j,0} := Q_{j-1,m_{j-1}}$ for $j \ge 1$. If

$$m_j \ \ge\ \left\lceil \frac{\log(s_j/\eta_j)}{\log(1/\rho_j)} \right\rceil, \qquad j = 0, \ldots, J, \tag{20}$$

then each stage remains in its local basin and

$$\|Q_{j,m_j} - \bar{Q}^{\dagger}_{\tau_j}\|_{2,\bar{\mu}} \le \eta_j, \qquad j = 0, \ldots, J.$$

In particular, the final iterate satisfies

$$\|Q_{J,m_J} - \bar{Q}^{\dagger}_{\tau_J}\|_{2,\bar{\mu}} \le \eta_J.$$

Proof. The proof is by induction over the temperature grid. The assumed initialization places $Q_{0,0}$ in the first local basin. The stated stagewise contraction assumption and the choice of m_0 imply $\|Q_{0,m_0} - \bar{Q}^{\dagger}_{\tau_0}\|_{2,\bar{\mu}} \le \eta_0$. If the conclusion holds at stage j, then

$$\|Q_{j,m_j} - \bar{Q}^{\dagger}_{\tau_{j+1}}\|_{2,\bar{\mu}} \le \eta_j + \|\bar{Q}^{\dagger}_{\tau_j} - \bar{Q}^{\dagger}_{\tau_{j+1}}\|_{2,\bar{\mu}} \le s_{j+1},$$

by (19). Hence the warm start for stage $j+1$ lies in the next basin, and another application of the stagewise contraction gives the desired η_{j+1} error after m_{j+1} iterations. Repeating this argument proves the claim through temperature τ_J. $\square$

Corollary 3 (Conditional hard-limit annealing)**.** *Suppose the finite-schedule assumptions of Theorem 7 hold along every finite prefix of an infinite grid $\tau_j \downarrow 0$ in the fixed reference norm $\|\cdot\|_{2,\bar{\mu}}$. Assume additionally that the path of fixed-reference projected fixed points is continuous at the hard limit,*

$$\bar{Q}^{\dagger}_{\tau_j} \longrightarrow \bar{Q}^{\dagger}_{\mathrm{hard}},$$

and that the basin-overlap condition can be maintained with tolerances $\eta_j \downarrow 0$. Then the annealed population iterates converge to $\bar{Q}^{\dagger}_{\mathrm{hard}}$.

Proof. Theorem 7 gives terminal stage errors bounded by η_j. The triangle inequality gives

$$\|Q_{j,m_j} - \bar{Q}^{\dagger}_{\mathrm{hard}}\|_{2,\bar{\mu}} \le \eta_j + \|\bar{Q}^{\dagger}_{\tau_j} - \bar{Q}^{\dagger}_{\mathrm{hard}}\|_{2,\bar{\mu}},$$

and both terms vanish by assumption. $\square$

Further continuation strategies. One could also vary the discount factor or combine discount and temperature continuation. We do not formalize that extension here: it would require its own common-norm stability and basin-overlap assumptions, and is separate from the population temperature-annealing guarantee above.

H.5 Gradient-descent interpretation

A useful perspective on the projected soft Bellman update is obtained by writing it in residual form. Define

$$F_{\mathcal{F}}(Q) := Q - \mathcal{T}_{\mathcal{F}}(Q),$$

so the population iteration becomes

$$Q^{(k+1)} = Q^{(k)} - F_{\mathcal{F}}(Q^{(k)}).$$

This is a unit-step projected gradient method, and its behavior near $Q^\star$ is governed by the local geometry of $F_{\mathcal{F}}$.

Suppose for simplicity that $Q^\star$ lies in the interior of $\mathcal{F}$, so that the projection is smooth in a neighborhood of $Q^\star$. At the soft-optimal fixed point $Q^\star$, the Jacobian of $F_{\mathcal{F}}$ is

$$DF_{\mathcal{F}}(Q^\star) = I - \gamma\, \Pi_{\mathcal{F}}\, T_{\pi^\star}^{\mathrm{eval}}.$$

Because $T_{\pi^\star}^{\mathrm{eval}}$ is a γ-contraction in the stationary norm and $\Pi_{\mathcal{F}}$ is nonexpansive in this geometry, the operator $\gamma\, \Pi_{\mathcal{F}} T_{\pi^\star}^{\mathrm{eval}}$ has norm at most γ. Consequently, $DF_{\mathcal{F}}(Q^\star)$ is positive definite on the action-differential subspace, implying that $F_{\mathcal{F}}$ is locally *strongly monotone* and smooth in a neighborhood of $Q^\star$ [7]. In this region, the projected soft Bellman update behaves like gradient descent on a locally strongly convex, smooth objective.

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   Answer: [Yes].

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